

Asymmetric Information in Frictional Markets for Liquidity: Collateralized Credit vs Asset Sale*

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Abstract

This paper studies (non-)equivalence of collateralized credit and asset sales in over-the-counter markets subject to commitment and information frictions. Embedded in a search-theoretic general equilibrium model, a signaling game refined by the undefeated equilibrium endogenizes the choice between pooling and separating offers and addresses the payment puzzle. The results show that non-equivalence depends on economic fundamentals, commitment, and information frictions. Despite adverse selection, first-best consumption can occur for collateralized credit, but not for asset sales, with belief-driven endogenous haircuts and over-collateralization characterizing the terms of trade. An extension incorporating co-existing information-sensitive assets and fiat money sheds new light on portfolio management and diversification under private information and ties optimal payment strategies to monetary policy.

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1 Introduction

Asset sales and collateralized credit represent essential liquidation strategies to obtain funds in financial markets. Given the large array of asset classes traded, more often than not, such transactions are subject to information frictions, resulting in haircuts and over-collateralization.¹ As documented by [Plantin \(2009\)](#), it is commonly the holder of the asset that has more information about its quality than the counterpart.² Examples are vast, ranging from counterfeits of commodity money and private bank notes in historical episodes to corporate debt, collateralized debt obligations, and mortgage- and asset-backed securities in modern times.

Internalizing said characteristics, this paper presents a search-theoretic general equilibrium model to study optimal payment strategies in decentralized markets subject to commitment and information frictions. At its core, it challenges the allocation-equivalence of asset sales and collateralized credit raised in the *payment puzzle* by [Lagos \(2011\)](#) and provides novel insights under what conditions either payment strategy dominates the other. Rigorous game-theoretic foundations give rise to belief-driven debt limits, adding to the discussion on haircuts and over-collateralization, and provide guidance on the formation of risk-free contracts in credit markets plagued by asymmetric information. Incorporating co-existing information-sensitive assets and fiat money, the general equilibrium analysis further sheds new light on portfolio management and diversification under adverse selection and ties optimal payment strategies to monetary policy.

The model is deeply founded in the monetary search literature drawing from [Lagos and Wright \(2005\)](#) and [Rocheteau \(2011\)](#). There are two fixed types of agents – consumers and producers – trading in two sequentially opening goods markets – a centralized market (CM) and a decentralized market (DM). Media of exchange are essential to realizing gains from trade in the DM due to commitment and enforcement frictions, where asset sales and secured credit characterize the incentive-feasible settlement strategies. Under uncertainty regarding the stochastic terminal value, agents acquire a portfolio of assets (and in an extension information-insensitive fiat money) in the CM. Upon being privately informed about the idiosyncratic quality of the portfolio, consumers and producers bilaterally bargain in the DM, where a signaling game refined by the *undefeated equilibrium* by [Mailath et al. \(1993\)](#) characterizes the terms of trade. The refinement endogenously selects between pooling and separating offers by maximizing the surplus of consumers holding high-quality assets, thus eliminating Pareto inefficiencies.

¹A haircut is defined as the difference between the market value of an asset used as collateral and the effective size of the granted loan. See [Baklanova et al. \(2015\)](#), [Baklanova et al. \(2019\)](#), [Copeland et al. \(2012, 2014\)](#), [Copeland et al. \(2014a\)](#), [Dang et al. \(2013, 2015\)](#), [Gorton and Metrick \(2012a,b\)](#), [Gorton and Ordoñez \(2014\)](#), and [Krishnamurthy et al. \(2014\)](#) for recent discussions on haircuts and over-collateralization in over-the-counter derivatives.

²The ‘learning-by-holding’ approach of [Plantin \(2009\)](#) argues that holders of an asset learn quicker about its quality than other investors in the market, creating information frictions. Learning-by-holding is particularly relevant for privately-issued debt and collateralized debt obligations. Such assets are often not publicly traded, hindering the investors’ ability to fully diversify their portfolios. Furthermore, they provide investors with privileged access to information creating information frictions.

The results show that allocation-equivalence of collateralized credit and asset sales is guaranteed for separating offers, but not for pooling offers, where the (non-)equivalence hinges on the producers' ability to commit to return the pledged collateral upon repayment of the agreed-upon loan obligation. More specifically, the distinction between non-recourse (the producer is not committed) and recourse (the producer is committed) loans.³

Consider first a benchmark case in which the producer cannot commit to future actions (a non-recourse loan).⁴ In other words, producers can choose to refuse the consumers' repayment of the credit obligation and keep the pledged collateral instead, should they desire to do so. Given that premise, a consumer only proposes offers that make the producer indifferent between returning the collateral or not, while a producer exclusively accepts offers that yield a non-negative expected surplus. The consequence is allocation-equivalence of collateralized credit and asset sales despite information frictions, where the undefeated equilibrium, i.e., the selection between pooling and separating offers, depends on the distribution of asset qualities in the economy. If the share of high-quality assets is low, the equilibrium is separating, while an abundance of high-quality assets renders a pooling offer the undefeated equilibrium. The results are in line with [Li and Rocheteau \(2008\)](#) and [Bajaj \(2018\)](#) applying the undefeated equilibrium to a monetary search model with indivisible assets.

Allocations change once the producer is committed to honoring the contract (a recourse loan), and thus returns the pledged collateral upon repayment of the agreed-upon loan obligation. In said scenario, a pooling sale represents a limiting case of a pooling credit, and thus every allocation attainable with an asset sale can be replicated with a collateralized credit, but not vice versa. The reason is intuitive and relies on the producer's beliefs about the quality of the traded asset and the consumers' default decision. While a pooling sale requires the asset to be priced at the expected market value, the haircut of a pooling credit only needs to successfully eliminate the consumers' incentive for strategic default. The effective size of the haircut is endogenous and depends on fundamentals. If the aggregate asset supply is sufficiently large, then the undefeated equilibrium is a pooling collateralized credit, where provided consumers hold enough assets to support the required over-collateralization, consumption can be first-best despite information frictions. If assets are relatively scarce, on the other hand, then an outright sale priced at the expected market value renders the undefeated strategy. Unlike in the previous studies by [Li and Rocheteau \(2008\)](#) and [Bajaj \(2018\)](#), if loans are recourse, then a separating equilibrium is defeated at all times given the inefficiencies arising from the signaling cost incurred through asset retention.

³Alternatively, instead of a distinction between 'not committed' and 'committed', one could envision differences in contractual enforcement to return the collateral to the borrower once credit obligations are met. For a discussion of non-recourse loans see [Geanakoplos \(1996\)](#), [Araujo et al. \(2000\)](#), [Geanakoplos and Zame \(2014\)](#), and [Maurin \(2022\)](#). Recourse loans, on the other hand, are studied in [Muley \(2015\)](#), [Kahn and Park \(2016\)](#), and [Gottardi et al. \(2019\)](#).

⁴Note that limited commitment by the producer is all but necessary for a monetary equilibrium to exist since in the absence of enforcement and record-keeping technologies only the consumer's inability to commit represents a necessary and sufficient condition for a medium of exchange to be essential. See the earlier work by [Kocherlakota \(1998\)](#), [Wallace \(2001\)](#), and [Shi \(2006\)](#) motivating the necessity of a medium of exchange in frictional goods markets.

The infinite-horizon structure of the framework further allows to study the co-existence of information-sensitive assets and fiat money, thus tying optimal payment strategies to monetary policy. Focusing on recourse loans, first and foremost, if the aggregate asset supply is large enough to support first-best allocations without money, fiat money is not valued in equilibrium unless monetary policy corresponds to the Friedman rule. If pure asset transactions yield constrained-efficient allocations, payment strategies vary with the nominal interest. At the Friedman rule, allocations are socially optimal. Since first-best allocations are attainable with combinations of fiat money and a pooling credit priced at the low-quality asset, the consumer has no preference to spend fiat money first, and thus there exists no pecking-order theory of payments despite information frictions. Away from the Friedman rule, there exists a positive threshold nominal interest rate below which transactions are settled using a combination of fiat money and a pooling collateralized credit, and above which transactions are settled using a combination of fiat money and a pooling sale. In both cases, fiat money represents the preferred medium of exchange since the constrained-efficient allocations require cross-subsidization of low-quality asset holders. The associated distortions give rise to a pecking-order theory of payment choices in line with [Rocheteau \(2011\)](#), [Lester et al. \(2012\)](#), and [Jacquet and Tan \(2012\)](#).

The rest of the paper is organized as follows. The remainder of Section 1 discusses the related literature. Section 2 introduces the baseline environment without information frictions, while Section 3 characterizes the baseline equilibrium. Section 4 introduces information frictions and solves the bargaining game under private information. The benchmark case studying 'non-recourse' loans is presented in Subsection 4.1. The co-existence of fiat money and information-sensitive assets, as well as the optimal monetary policy under adverse selection, are discussed in Section 5. Lastly, Section 6 concludes.

1.1 Related Literature

First and foremost, this paper contributes to the broad literature analyzing collateralized loans and repurchase agreements. Among the first to show that collateral allows agents to overcome solvency concerns was [Bester \(1985\)](#), later revisited by [Kiyotaki and Moore \(1997\)](#) in a search environment subject to commitment and enforcement frictions. Further general equilibrium frameworks modeling collateralized loans to overcome commitment and solvency concerns involve [Geanakoplos \(1996, 2002\)](#), [Araujo et al. \(2000\)](#), [Simsek \(2013\)](#), [Geanakoplos and Zame \(2014\)](#), and [Antinolfi et al. \(2015\)](#). [Flannery \(1996\)](#), in turn, argues that while market participants may know about the solvency of the respective counterparty, they may not be aware of the quality of individual assets in their portfolio. An idea closely related to [Gorton and Ordoñez \(2014\)](#), [Ozdenoren et al. \(2021b\)](#), and the paper at hand, with the crucial difference that in [Flannery \(1996\)](#) private credit collapses due to an [Akerlof \(1970\)](#) lemons problem rather than giving rise to endogenous haircuts.

Collateralized loans featuring endogenous information frictions, in turn, are studied by [Dang et al. \(2013, 2015\)](#) and [Yang \(2019\)](#) promoting the benefits of information-insensitive debt in line with the aforementioned papers and the results in this study.⁵ For a dynamic approach to asset-backed securities involving long-lived information-sensitive assets see [Ozdenoren et al. \(2021a\)](#).

Next to the existing literature on collateralized loans, this paper furthermore contributes to the lineage of research by [Duffie \(1996\)](#), [Kehoe and Levine \(2001\)](#), [Lacker \(2001\)](#), [Rampini \(2005\)](#), and [Ferraris and Watanabe \(2008\)](#) in which collateralized credit emerges endogenously. Abstaining from information frictions, [Berentsen and Waller \(2011\)](#) conclude that any allocation in an economy with asset sales can be replicated by an economy with collateralized credit, but not vice versa. Introducing heterogeneous valuations for the traded assets, [Mills and Reed \(2012\)](#) study strategic default and provide an optimal contract in the spirit of [Kehoe and Levine \(1993\)](#) by analyzing varying degrees of commitment on the borrower and the lender's side in line with the paper at hand.⁶ However, contrary to this paper, idiosyncratic valuations for the asset are public information precluding an analysis of the endogenous selection between secured loans and outright sales when haircuts are belief-driven.⁷ Focusing on search frictions, [Tomura \(2016\)](#) explains the demand for collateralized credit with a hold-up problem introduced through a deadline for retrieving cash.⁸ [Monnet and Narajabad \(2017\)](#), in turn, suggest non-equivalence relying on a borrower's uncertainty regarding the value of holding the security in the future. [Awaya et al. \(2021\)](#) revisit this logic and show that if agents have no option to re-acquire the collateral prior to the next trade, the endogenous borrowing limit can be larger than the intrinsic value of the asset. [Parlatore \(2019\)](#) studies the trade-off between collateralized credit and asset sales under uncertainty regarding the outcome of a risky project. In her model, an increase in liquidity increases the liquidity premium of assets, while the collateral premium declines, suggesting non-equivalence. [Gottardi et al. \(2019\)](#), on the other hand, study the role of repurchase agreements as an insurance against price fluctuations focusing on the aspect of re-hypothecation.

The promoted payment strategies also borrow elements from the literature on asset prices and private information, pioneered by [Akerlof \(1970\)](#), as well as the extensive security design literature including [Leland and Pyle \(1977\)](#), [Boot and Thakor \(1993\)](#), [Myers and Majluf \(1984\)](#), [Nachman and Noe \(1994\)](#), [DeMarzo and Duffie \(1995\)](#), [Demarzo and Duffie \(1999\)](#), [Biais et al. \(2007\)](#), and [Chemla and Hennessy \(2014\)](#). By embedding game-theoretic foundations into a monetary general equilibrium model, this paper further contributes to the existing research analyzing the effects

⁵See [Kurlat \(2013\)](#) and [Bigio \(2015\)](#) for an application to production economies subject to information frictions.

⁶In doing so, the paper is in line with the large literature on asset reallocation in over-the-counter markets including [Duffie et al. \(2005\)](#), [Vayanos and Wang \(2007\)](#), [Lagos and Rocheteau \(2009\)](#), [Lagos et al. \(2011\)](#), [Camargo and Lester \(2014\)](#), [Chiu and Koepl \(2016\)](#), [Madison \(2019\)](#), and [Wang \(2020\)](#).

⁷A recent paper by [Kang \(2021\)](#) also considers the role of information frictions to explain the non-equivalence of asset sales and collateralized credit in a search model, but refrains from explicitly modeling a bargaining problem under private information.

⁸The paper is in line with [Martin et al. \(2014\)](#) analyzing the U.S. tri-party repo market arguing that in the presence of information frictions, the repo market can collapse due to a coordination failure among investors. Contrary to [Martin et al. \(2014\)](#), [Tomura \(2016\)](#) abstains from information frictions confirming that excluding information-sensitive assets does not preclude a fragile repo market.

of information frictions on the moneyiness of an asset in line with [Lester et al. \(2012\)](#), [Li et al. \(2012\)](#), and [Jacquet and Tan \(2012\)](#). The first to study the effects of information frictions in a search-theoretic model using pooling equilibria were [Velde et al. \(1999\)](#). Separating equilibria relying on the [Cho and Kreps \(1987\)](#) Intuitive Criterion, in turn, were studied by [Nosal and Wallace \(2007\)](#), [Guerrieri et al. \(2010\)](#), [Rocheteau \(2011\)](#), and [Madison \(2019\)](#), among others. To endogenize the selection of pooling and separating equilibria, a first application of the [Mailath et al. \(1993\)](#) undefeated equilibrium, using a [Shi \(1995\)](#) and [Trejos and Wright \(1995\)](#) second-generation monetary search model with indivisible assets, was provided by [Li and Rocheteau \(2008\)](#), followed by [Bajaj \(2018\)](#). Explicitly modeled distinctions between collateralized credit and asset sales, as well as variations in commitment frictions, however, remain absent. For an application of the undefeated equilibrium to asset reallocation with two-dimensional private information and partially-liquid assets see [Wang \(2020\)](#).

2 Baseline Environment

Time is discrete, starts at $t = 0$, and continues forever.⁹ Each period is divided into two subperiods: a centralized market (CM) and a decentralized market (DM). While exchange in the CM occurs in a competitive Arrow-Debreu fashion, in the DM agents meet bilaterally at random and bargain over the terms of trade. There is a unit measure of two distinct types of agents, consumers and producers, named after their preferences in the DM, and denoted by the subscripts $i = \{c, p\}$. While both types can produce and consume in the CM, in the DM, consumers want to consume but cannot produce and producers can produce but do not want to consume, generating gains from trade. The expected lifetime utility of an agent $i = \{c, p\}$ is:

$$\sum_{t=0}^{\infty} \beta^t \mathcal{U}_i(q_t, z_t, h_t), \quad (1)$$

with $\mathcal{U}_c(q_t, z_t, h_t) \equiv u(q_t) + z_t - h_t$ and $\mathcal{U}_p(q_t, z_t, h_t) \equiv -q_t + z_t - h_t$, where $q_t \in \mathbb{R}_+$ denotes the non-durable consumption good in the DM, $z_t \in \mathbb{R}_+$ the non-durable consumption good in the CM, and $h_t \in \mathbb{R}_+$ hours worked in the CM. For tractability, the period utility is separable across subperiods. The utility of consumption in the DM, $u(q)$, is continuously differentiable, strictly increasing in q , and concave with $u'(q) > 0 > u''(q)$, $u(0) = 0$, $u'(0) = \infty$, and $u'(\infty) = 0$. The disutility of production, $-q$, is linear. Efficiency requires $q = q^*$, where q^* solves $u'(q^*) = 1$. Consumption and production in the CM are linear with labor being the only input. Finally, the discount factor across periods is $\beta \in (0, 1]$, where $\beta = (1 + r)^{-1}$ and $r \geq 0$ is the rate of time preference. In the following, the timing of events, visualized in Figure 1, is discussed in detail.

⁹The assumption of an infinite horizon is not strictly necessary for the analysis of short-lived assets only but allows for a comparison to standard monetary models. Further, it makes the framework readily extendable to incorporate long-lived assets – see Section 5 discussing the co-existence of fiat money and short-lived information-sensitive assets.

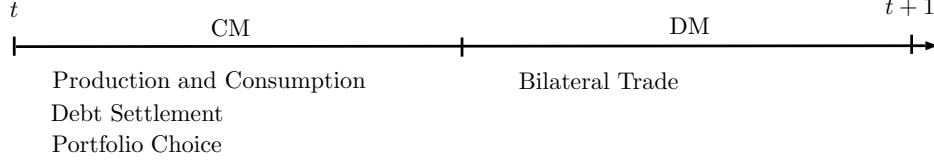


Figure 1: Timing of events: baseline environment

At the beginning of the CM, each agent is endowed with $A \in \mathbb{R}_+$ units of a perfectly divisible one-period lived real asset yielding $\kappa > 0$ units of the numéraire good to the bearer in the subsequent CM. Once asset endowments are received, agents produce and consume the numéraire good, z , adjust their asset holdings for the following DM by trading z units of the numéraire good for a units of the asset at the competitive price ψ , and settle credit obligations from the previous DM.

In the DM, consumers and producers meet bilaterally at random with probability one and remain matched until the beginning of the subsequent CM. There is no record-keeping of transactions and consumers cannot commit to future actions, eliminating unsecured credit. Given the non-durability of the numéraire good, a medium of exchange – the asset a – is essential for trade to occur. Two settlement options are incentive-feasible: asset sales and collateralized credit, denoted by the calligraphic subscripts $j = \{\mathcal{S}, \mathcal{C}\}$. In the event of an asset sale, transactions are settled instantaneously, where a consumer transfers $y_{\mathcal{S}} \in (0, a]$ units of the asset to the producer in return for q units of the DM good. In the case of a collateralized credit, a consumer pledges $y_{\mathcal{C}} \in (0, a]$ units of the asset in return for q units of the DM good, and agrees to repurchase the pledged asset in the subsequent CM at a price defined in the DM, where l represents the agreed-upon liability. Should a consumer default, the producer keeps the pledged asset. In the event of repayment, on the other hand, the redemption of the collateral is guaranteed.¹⁰

3 Baseline Equilibrium

Let us now determine the baseline equilibrium without information frictions. An agent, $i \in \{c, p\}$, enters the CM with a units of the real asset and liabilities, l , from the previous period to solve:

$$W_i(a, l; \kappa) = \max_{z, h, a'} z - h + V_i(a'; \kappa'), \quad (2)$$

$$\text{s.t. } z + \psi a' = h + \kappa a + \psi A - l, \quad (3)$$

where $V_i(a'; \kappa')$ denotes the continuation value of entering the DM with a' units of the asset. Agents finance end-of-period asset balances, a' , CM consumption, z , and outstanding credit obligations, l , through inputs of labor, h , and asset returns, κa , subject to the budget constraint (3). Since

¹⁰In subsection 4.1, this assumption is relaxed to study the benchmark case with non-recourse loans where redemption of the collateral is not guaranteed.

$W_i(a, l; \kappa)$ is linear in current wealth, an agent's future wealth is independent of current wealth as shown by the first-order and envelope conditions, $V_{a'} = \psi$, $W_a = \kappa$, and $W_l = -1$. The value functions of a consumer and a producer entering the DM with a units of the asset are:

$$V_c(a; \kappa) = u(q) + \beta W_c(a - y_S, l; \kappa), \quad (4)$$

$$V_p(a; \kappa) = -q + \beta W_c(a + y_S, l; \kappa). \quad (5)$$

Due to bilateral matching, and thus $q_c = q_p$, $y_{S,c} = y_{S,p}$, $y_{c,c} = y_{c,p}$, and $l_c = l_p$, the subscripts c and p are omitted from here on onward. Using the linearity of the CM value function, the consumers' and the producers' match surpluses can be reduced to $S_c(a; \kappa) = u(q) - \beta y_S \kappa - \beta l$ and $S_p(a; \kappa) = -q + \beta y_S \kappa + \beta l$, laying the foundation for the bargaining game in the DM.

The terms of trade, (q, y_S, y_C, l) , are determined by consumers' take-it-or-leave-it offers, where q denotes the amount of DM goods received, y_S the amount of assets sold, y_C the amount of assets pledged as collateral, and l the credit obligation in period $t + 1$. For simplicity, the consumers' asset holdings are assumed to be common knowledge in the match, characterizing the following bargaining problem:¹¹

$$(q, y_S, y_C, l) \in \arg \max_{q, y_S, y_C, l} u(q) - \beta y_S \kappa - \beta l, \quad (6)$$

$$\text{s.t. } -q + \beta y_S \kappa + \beta l \geq 0, \quad (7)$$

$$\text{s.t. } -l + y_C \kappa \geq 0, \quad (8)$$

$$\text{s.t. } -y_S - y_C + a \geq 0, \quad (9)$$

where (7) represents the producer's participation constraint, (8) the collateral constraint, and (9) the feasibility constraint.¹² Combining the constraints and incorporating the fact that (7) holds with equality given the terms of trade, the consumer's problem reduces to:

$$\max_q u(q) - q \quad \text{s.t. } -q + \beta a \kappa \geq 0, \quad (10)$$

yielding the DM allocations $q = \min\{q^*, \beta a \kappa\}$, $y_j = \min\{q/\beta \kappa, a\}$, and $l = y_C \kappa$, where $q = q^*$ if the consumer's financial wealth is sufficient to cover the producer's disutility of production, and $q < q^*$ otherwise. In line with Lagos (2011), in the absence of information frictions, collateralized credit and asset sales are allocation-equivalent.

Let us now incorporate the solutions of the bargaining game into the general equilibrium structure of the model to determine an agent's asset holdings, a , and the equilibrium asset price,

¹¹This allows to avoid specifying a producer's beliefs regarding the consumer's asset holdings. The assumption is without loss of generality since a consumer's surplus is weakly monotonically increasing in her asset holdings, and hence if a consumer had the possibility to report her asset holdings prior to the match there would be an equilibrium in which she would do so truthfully.

¹²Note from (8) that in the absence of information frictions debt limits are imposed exogenously following Kiyotaki and Moore (1997).

ψ . Plugging (3), (4), and (5) into (2), an agent's problem reduces to:

$$a_i \in \arg \max_a -(\psi - \beta\kappa)a + S_i(a; \kappa) \geq 0. \quad (11)$$

Given (11), an agent chooses a to maximize her DM surplus, $S_i(a; \kappa)$, net of the cost of holding assets, $-(\psi - \beta\kappa)a$. If $\psi > \beta\kappa$, assets are costly to hold and there is a unique solution to (11). For $\psi < \beta\kappa$, however, demand is infinite, and hence there is no solution. Therefore, in equilibrium, the latter case is ruled out, assuring $\psi \geq \beta\kappa$.

Definition A. *An equilibrium is a list of asset holdings, a , terms of trade in the DM, (q, y_S, y_C, l) , and the aggregate asset supply, A , such that a solves (11), (q, y_S, y_C, l) solves (6)-(9), and the market clearing condition, $\int_{j \in N} a(j) dj = A$, holds with N being the total number of agents in the economy.*

There exists a unique solution to (11) yielding the equilibrium CM asset price $\psi = \beta\kappa[1 + \mathcal{L}]$, where $\beta\kappa$ corresponds to the discounted fundamental value and $\mathcal{L} \equiv u'(q) - 1$ to the liquidity premium. If $q < q^*$, then $\mathcal{L} > 0$, and thus $\psi > \beta\kappa$ since consumers would prefer an additional unit of the asset in the DM. If $q = q^*$, on the other hand, then $\mathcal{L} = 0$ and assets trade at the fundamental value $\psi = \beta\kappa$ in the CM. Note that since producers have no use for the asset in the DM, and thus $a_p = 0$, consumers are the only ones accumulating assets in the CM, and each consumer enters the DM with the same amount of assets, a .

4 Bargaining under Private Information

Having determined the baseline equilibrium of the model, we now alter the environment to study the bargaining game under private information. At the beginning of the DM, after assets are accumulated, agents are privately informed about the idiosyncratic quality of their portfolio. The realization is common to all assets held by an agent but independent across agents and all consumers enter the DM with the same asset holdings, a . Furthermore, the timing of events, as visualized in Figure 2, implies that once informed, consumers cannot reallocate/ diversify their portfolio prior to the bargaining game in the DM.¹³

Assume asset returns are stochastic with $\kappa^X = \{\kappa^L, \kappa^H\}$. With probability $\pi \in (0, 1)$, the asset yields a high terminal return, κ^H , while with the complementary probability, $1 - \pi$, the return is low, satisfying:

$$\kappa^X = \begin{cases} \kappa^H & \text{with probability } \pi \in (0, 1) \\ \kappa^L & \text{with probability } 1 - \pi \in (0, 1), \end{cases} \quad (12)$$

¹³See e.g. [Madison \(2019\)](#) for a monetary framework studying asset reallocation in over-the-counter markets under adverse selection.

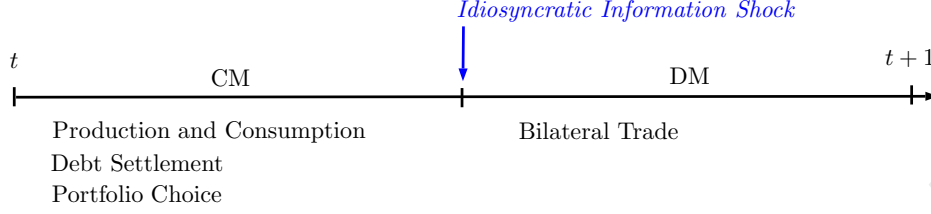


Figure 2: Timing of events: bargaining under private information

with $0 \leq \kappa^L < \kappa^H < \infty$, where $R \equiv \pi\kappa^H + (1 - \pi)\kappa^L$ denotes the expected return. The bargaining game has the structure of a signaling game, i.e., the informed agent moves first and makes the offer.¹⁴ A strategy for the consumer is to specify an offer $(q, y_S, y_C, l) \in \mathcal{F} \equiv \mathbb{R}_+ \times [0, a]$ as a function of her type, κ^χ , and her asset holdings, a . A strategy for the producer, on the other hand, is to form expectations about the terminal return of the consumer's asset, κ^χ , and determine an acceptance rule that specifies the set $\mathcal{A}_p(\lambda) \subseteq \mathcal{F}$ of acceptable offers.¹⁵ Let $\lambda(\kappa^\chi \mid (q, y_S, y_C, l)) \in [0, 1]$ represent the producer's posterior belief that the consumer's asset is of quality $\chi = \{L, H\}$ conditional on the offer, (q, y_S, y_C, l) , made. The posterior expected return can therefore be formulated as:

$$\mathbb{E}_\lambda[\kappa^\chi] = \sum_{\chi=L,H} \lambda(\kappa^\chi \mid (q, y_S, y_C, l)) \kappa^\chi, \quad (13)$$

determining the producer's set of acceptable offers:

$$\mathcal{A}_p(\lambda) = \{(q, y_S, y_C, l) \in \mathcal{F} : -q + \beta y_S \mathbb{E}_\lambda[\kappa^\chi] + \beta l \geq 0 \text{ and } -l + y_C \mathbb{E}_\lambda[\kappa^\chi] \geq 0\}. \quad (14)$$

Hence, for a given belief system, $\lambda(\kappa^\chi \mid (q, y_S, y_C, l))$, in order for a producer to accept an offer made by the consumer, the offer has to yield a non-negative expected surplus. Furthermore, debt limits are determined endogenously from the producer's posterior belief about the quality of the traded asset and the consumer's default decision. Assuming a tie-breaking rule according to which a producer agrees to any offer that makes her indifferent between accepting and rejecting, a κ^χ -consumer solves:

$$(q^\chi, y_S^\chi, y_C^\chi, l^\chi) \in \arg \max_{q, y_S, y_C, l} [u(q) - \beta y_S \kappa^\chi - \beta y_C f] \mathbb{I}_{\mathcal{A}_p} \quad \text{s.t. } (q, y_S, y_C, l) \in \mathcal{F}, \quad (15)$$

where $f \in [\kappa^L, \kappa^H]$ denotes the face value of the debt and $\mathbb{I}_{\mathcal{A}_p}$ is an indicator function equal to one if the consumer's offer is in the producer's set of acceptable offers, i.e., if $(q, y_S, y_C, l) \in \mathcal{A}_p(\lambda)$, and zero otherwise. Precisely, a κ^χ -consumer proposes an offer $(q, y_S, y_C, l) \in \mathbb{R}_+ \times [0, a]$ to maximize

¹⁴I abstain from a separate discussion of the bargaining game with heterogeneous asset returns under complete information, as it is equivalent to (6)-(9) with $\kappa^\chi = \{\kappa^L, \kappa^H\}$.

¹⁵To test the sensitivity of the results to the bargaining protocol, a formal analysis of the model using an alternative screening game with producers' take-it-or-leave-it offers is provided in the Supplementary Appendix K. Albeit quantitatively different, the qualitative results are in line with the results of the discussed signaling game.

her surplus in the DM, taking as given the producer's acceptance rule, $\mathcal{A}_p(\lambda)$. Let us define the equilibrium in the DM.

Definition B. *An equilibrium of the bargaining game is a system of beliefs, $\lambda(\kappa^x \mid (q, y_S, y_C, l))$, a set of equilibrium offers, $(q^x, y_S^x, y_C^x, l^x) \in \mathcal{O}$, and an acceptance rule, $\mathcal{A}_p(\lambda)$, such that:*

- (i) *Posterior beliefs, $\lambda(\kappa^x \mid (q, y_S, y_C, l))$, are derived from Bayes' rule whenever possible;*
- (ii) *Conditional on $\lambda(\kappa^x \mid (q, y_S, y_C, l))$, the strategies $(q^x, y_S^x, y_C^x, l^x) \in \mathcal{O}$ and $\mathcal{A}_p(\lambda)$ are sequentially rational; and*
- (iii) *The equilibrium is undefeated.*

The first two conditions, (i) and (ii), define a Perfect Bayesian Equilibrium (PBE), where an equilibrium for this bargaining game is a profile of strategies for the consumer and the producer, and a system of beliefs. If (q, y_S, y_C, l) is an offer made by the consumer, then $\lambda(\kappa^x \mid (q, y_S, y_C, l))$ is derived from the producer's prior belief according to Bayes' rule whenever possible. A formal definition of the belief system is provided in Appendix A. However, note that both conditions (i) and (ii) do not impose restrictions on out-of-equilibrium beliefs. To obtain reasonable results, condition (iii) applies the undefeated equilibrium refinement by Mailath et al. (1993).¹⁶ The refinement endogenously selects between separating and pooling offers, i.e., $(q^L, y_S^L, y_C^L, l^L) \neq (q^H, y_S^H, y_C^H, l^H)$ and $(q^L, y_S^L, y_C^L, l^L) = (q^H, y_S^H, y_C^H, l^H) = (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})$. Precisely, a PBE (A) is defeated by another PBE (B) if there exists an offer that is not made by any consumer in A but made by a set of consumers Ω in B and all consumers in Ω are weakly better off than the consumers in A, while some consumers in Ω are strictly better off. In Proposition C it is shown that an equilibrium is undefeated if it maximizes the high-type consumer's surplus, $S_c(a; \kappa^H)$, and if there exists no other equilibrium offer such that either the high-type consumer has an incentive to unilaterally deviate to a separating offer (i.e., $\Omega = H$), or both types prefer a deviation to a pooling offer (i.e., $\Omega = \{L, H\}$). This includes the choice between collateralized credit and asset sale. A formal definition of the undefeated equilibrium is provided in Appendix B.

Let us start by characterizing the least-inefficient separating offers, $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H)\} \in \mathcal{O}$.

Proposition A. (Separating offers) *For $\kappa^L > 0$, the unique separating offers, $\{(q^L, y_S^L, y_C^L, l^L),$*

¹⁶The undefeated equilibrium has several advantages to the Intuitive Criterion by Cho and Kreps (1987) frequently used in the literature: (1) analog to the Intuitive Criterion it selects the least-inefficient separating equilibrium among all separating equilibria, (2) it allows for pooling equilibria thus guaranteeing a solution for $\kappa^L = 0$, and (3) it selects the PBE maximizing the surplus of the high-type consumers ensuring Pareto optimality.

$(q^H, y_S^H, y_C^H, l^H) \in \mathcal{O}$, are:

$$q^L = \min\{q^*, \beta a \kappa^L\}, \quad (16)$$

$$y_j^L = \min\{q^*/\beta \kappa^L, a\}, \quad (17)$$

$$l^L = y_C^L \kappa^L, \quad (18)$$

and:

$$q^H = [\kappa^H / \kappa^L] [u(q^H) - [u(q^L) - q^L]] \in (0, q^L), \quad (19)$$

$$y_j^H = q^H / \beta \kappa^H \in (0, a), \quad (20)$$

$$l^H = y_C^H \kappa^H, \quad (21)$$

with the corresponding comparative statics in Table 1. If $\kappa^L = 0$, then $q^X = y_j^X = l^X = 0$.

Proof. See Appendix C.

Proposition A shows that for the unique separating offers, a collateralized credit and an asset sale are allocation-equivalent. By signaling her true type, a consumer eliminates the producer's uncertainty regarding the future return of the provided asset, i.e., $\lambda(\kappa^L \mid (q^L, y_S^L, y_C^L, l^L)) = 1$ and $\lambda(\kappa^H \mid (q^H, y_S^H, y_C^H, l^H)) = 1$, allowing future credit obligations to be priced at the assets' fundamental value, i.e., $l^X = y_C^X \kappa^X$. Consider first a κ^L -consumer. A low-type can do no worse than under symmetric information, since this offer is always acceptable to the producer independent of their beliefs. At the same time, a low-type cannot do better, since every deviation would require pooling with a high-type's offer. Such out-of-equilibrium offers, $(q^o, y_S^o, y_C^o, l^o) \notin \mathcal{O}$, are ruled out by the binding incentive-compatibility constraint, suggesting $\lambda(\kappa^L \mid (q^o, y_S^o, y_C^o, l^o)) = 1$. Efficiency occurs for $\beta a \kappa^L \geq q^*$, and else $q^L < q^*$. Taking the low-type consumer's offer as a benchmark, a high-type consumer separates to guarantee $\lambda(\kappa^H \mid (q^H, y_S^H, y_C^H, l^H)) = 1$. This separation comes at a cost, where (19) uniquely determines $q^H \in [0, q^L]$. Given $q^H < q^L$, and thus $y_j^H \kappa^H < y_j^L \kappa^L$, it immediately follows that $y_j^H < y_j^L \in [0, a]$.¹⁷ In words, separation takes place through *asset retention*. The amount of assets retained is at the threshold such that the low-type has no incentive to mimic the high-type's offer – an application of Gresham's law suggesting that low-quality assets crowd out high-quality assets. In the limiting case with $\kappa^L = 0$, there is no trade, and thus $q^X = y_j^X = l^X = 0$.

Focusing on the κ^H -consumer, the comparative statics in Table 1 show that the high-type's consumption, q^H , and the amount of assets sold or pledged, y_j^H , decrease with an increase in the distance between the asset returns, i.e., $\partial q^H / \partial(\kappa^H - \kappa^L) < 0$ and $\partial y_j^H / \partial(\kappa^H - \kappa^L) < 0$, for any $q^L \in (0, q^*]$. Hence, the more severe the return heterogeneity, the more binding the incentive-

¹⁷Note that under full-information, since $0 \leq \kappa^L < \kappa^H < \infty$, the high-type consumes a weakly higher quantity than the low-type, $q^H \geq q^L \geq 0$, and spends a lower fraction of her assets, $y_j^H < y_j^L \in [0, a]$. To verify, solve (6)-(9) with $\kappa^X = \{\kappa^L, \kappa^H\}$.

	$q^L \in (0, q^*)$			$q^L = q^*$		
	κ^L	κ^H	a	κ^L	κ^H	a
$\partial q^L / \partial \cdot$	+	0	+	0	0	0
$\partial y_j^L / \partial \cdot$	0	0	+	-	0	0
$\partial q^H / \partial \cdot$	+	-	+	+	-	0
$\partial y_j^H / \partial \cdot$	+	-	(0, 1)	+	-	0

Table 1: Comparative statics: separating offers

compatibility constraint (19), and thus the lower q^H and y_j^H . The effect of a change in the consumer's asset holdings, a , on q^H and y^H requires a distinction between $q^L < q^*$ and $q^L = q^*$. If $q^L < q^*$, an additional unit of the asset raises the low-type consumer's surplus, i.e., $\partial q^L / \partial a > 0$, and thus the high-type consumer's marginal willingness to sell or pledge an additional unit of the asset is positive. However, given the binding incentive-compatibility constraint (19), a high-type can only use a fraction of each additional asset without generating the incentive for the low-type to mimic her offer, and thus $\partial y^H / \partial a \in (0, 1)$. If $q^L = q^*$, in turn, an additional unit does not relax the high-type's constraint, i.e., $\partial q^H / \partial a = 0$, and thus her willingness to sell or pledge another unit of the asset is zero, $\partial y^H / \partial a = 0$.

Having determined the least-inefficient separating offers, let us now determine the most-efficient pooling offer, $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \mathcal{O}$.

Proposition B. (Pooling offer) *The unique pooling offer, $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \mathcal{O}$, is characterized by:*

$$\bar{q} = u'^{-1}(f / [\pi f + (1 - \pi)\kappa^L]) \in (0, q^*), \quad (22)$$

$$\bar{y}_C = \bar{q} / \beta[\pi f + (1 - \pi)\kappa^L] \in (0, a], \quad (23)$$

$$\bar{l} = \bar{y}_C[\pi f + (1 - \pi)\kappa^L], \quad (24)$$

with the face value $f \in [\kappa^L, \kappa^H]$ satisfying $\partial f / \partial a \leq 0$ according to:

$$f \begin{cases} = \kappa^L & \text{if } a \geq a^* \\ \in (\kappa^L, \kappa^H) & \text{if } a \in (\tilde{a}, a^*) \\ = \kappa^H & \text{if } a \in (0, \tilde{a}], \end{cases} \quad (25)$$

and $\bar{y}_S = \{0, \bar{y}_C\}$ for $a \in \{(\tilde{a}, a^*), (0, \tilde{a}]\}$, where a^* solves $u'(\beta a^* \kappa^L) = 1$. The comparative statics are provided in Table 2.

Proof. See Appendix D.

Proposition B shows that conditional on the consumers' asset holdings, a , allocations for

	$\bar{q} \in (0, \tilde{q}]$			$\bar{q} \in (\tilde{q}, q^*)$			$\bar{q} = q^*$		
	κ^L	κ^H	π	κ^L	f	π	κ^L	κ^H	π
$\partial \bar{q} / \partial \cdot$	+	-	+	+	-	+	0	0	0
$\partial \bar{y}_S / \partial \cdot$	+/-	-	+/-	0	0	0	0	0	0
$\partial \bar{y}_C / \partial \cdot$	+/-	-	+/-	+/-	-	+/-	-	0	0

Table 2: Comparative statics: pooling offer

collateralized credit and asset sales can be non-equivalent in the unique pooling offer. More precisely, any allocation attainable with a pooling sale can be replicated by a pooling credit but not vice versa.

Consider a collateralized credit. Given the consumers' inability to commit to future actions, a producer expects low-type consumers to default on their credit obligation if the face value of the loan exceeds the fundamental value, i.e., if $f > \kappa^L$. High-type consumers, on the other hand, are expected to honor their obligation as long as $f \leq \kappa^H$ (with indifference under equality) due to the producer's commitment to return the high-quality asset upon repayment. Thus, to ensure non-negative expected profits, the producers' participation constraint needs to satisfy (23). Three cases are considered: (i) $f = \kappa^L$, (ii) $f \in (\kappa^L, \kappa^H)$, and (iii) $f = \kappa^H$. In case (i) the face value corresponds to the low return creating an information-insensitive contract. The result is over-collateralization resulting in first-best consumption $\bar{q} = q^*$ if $\beta a \kappa^L \geq q^*$ for both low- and high-type consumers, equivalent to the low-type consumer's complete information offer in (16)-(18). Thus, intuitively, a collateralized credit allows high-type consumers to mimic the low-type consumers' full-information offer and obtain first-best consumption if in possession of sufficient real balances, i.e., if $a \geq a^*$ with a^* solving $q^* = \beta a^* \kappa^L$, understanding that upon repayment of the credit obligation, they can consume the high return of their assets without the necessity to sacrifice consumption in the DM. Given the binding incentive compatibility constraint (19), this outcome is unattainable with a separating offer characterized in Proposition A. Once $a < a^*$, the high-type consumers' best response is to increase the face value, i.e., $\partial f / \partial a < 0$, with $f \in (\kappa^L, \kappa^H)$ for $a \in (\tilde{a}, a^*)$ (case (ii)). An immediate consequence is reduced demand, $\bar{q} < q^*$, since with every unit of the asset pledged a high-type consumer cross-subsidizes the consumption of the low-type consumer. In the limiting case with $a \in (0, \tilde{a}]$ (case (iii)), the face value corresponds to the return of the high-quality asset, i.e., $f = \kappa^H$, yielding allocation-equivalence to a pooling sale priced at the expected value $R \equiv \pi \kappa^H + (1 - \pi) \kappa^L$. In other words, a pooling sale represents the limiting case of a pooling credit with a binding liquidity constraint, i.e., with $a \in (0, \tilde{a}]$. Note that in all three cases no information is revealed and the producer's updated belief corresponds to the initial belief, i.e., $\lambda(\kappa^H | (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})) = \pi$ and $\lambda(\kappa^L | (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})) = 1 - \pi$.

The comparative statics in Table 2 distinguish between the three regimes (i) $a \geq a^*$, (ii) $a \in (\tilde{a}, a^*)$, and (iii) $a \in (0, \tilde{a}]$, yielding the corresponding DM quantities $\bar{q} = q^*$, $\bar{q} \in (\tilde{q}, q^*)$, and $\bar{q} \in (0, \tilde{q}]$, respectively. Starting with case (iii), if $a \leq \tilde{a}$, then the face value is $f = \kappa^H$. In

said case, an increase in the return heterogeneity (the share of high-types) decreases (increases) DM consumption, i.e., $\partial \bar{q}/\partial(\kappa^H - \kappa^L) < 0$ and $\partial \bar{q}/\partial \pi > 0$, since a decrease (increase) in κ^H (κ^L and π) reduces the κ^H -consumer's cross-subsidization of the κ^L -consumer. While the amount of assets sold decreases with an increase in the return of the high-quality asset, $\partial \bar{y}_j/\partial \kappa^H < 0$, the effects of an increase in the return of the low-quality asset, κ^L , or the share of high-quality assets, π , are ambiguous, with $\partial \bar{y}_j/\partial \kappa^L > 0$ and $\partial \bar{y}_j/\partial \pi > 0$ if and only if $\kappa^H > |Ru''(\bar{q})\bar{q}|$, where $\partial |u''(\bar{q})\bar{q}|/\partial \bar{q} < 0$. The comparative statics are qualitatively identical for case (ii) except for two key differences. First, if $a > \tilde{a}$, then $y_S = 0$ since κ^H -consumers strictly prefer a collateralized credit over an asset sale, and second, the face value $f < \kappa^H$ is strictly smaller than κ^H with $\partial f/\partial a \leq 0$ and $f = \kappa^L$ in the limiting case with $a = a^*$. Said limiting case represents case (i) with $\bar{q} = q^L = \min\{q^*, \beta a \kappa^L\}$ analog to the allocations of the low-type in the separating equilibrium.

Having determined the least-inefficient separating offers and the most-efficient pooling offer, let us now revisit (iii) in Definition B, and hence the selection of the undefeated equilibrium. Proposition C summarizes.

Proposition C. (Undefeated equilibrium) *The undefeated equilibrium satisfying Definition B is characterized by the pooling offer $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \mathcal{O}$ in (22)-(24).*

Proof. See Appendix E.

Proposition C shows that the undefeated equilibrium is characterized by the pooling offer $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \mathcal{O}$ defined in Proposition B for any $a \in (0, a^*]$ with a^* solving $q^* = \beta a^* \kappa^L$. Starting with $a \geq a^*$, if the consumer's liquidity constraint is non-binding, then the undefeated equilibrium is a pooling credit with the face value $f = \kappa^L$. In words, by offering a collateralized credit, a κ^H -consumer can attain the first-best allocation $\bar{q} = q^*$ without facing costly asset retention or cross-subsidization. A κ^L -consumer in turn would prefer to sell her assets at the pooling price, R , to exploit the information friction at the cost of the κ^H -consumer. However, given the belief system in Definition B, producers updating their beliefs understand that κ^H -consumers would never choose a face value $f > \kappa^L$ for $a > a^*$, and therefore attribute such out-of-equilibrium offers to κ^L -consumers revealing their type, i.e., $\lambda(\kappa^L \mid (q^o, y_S^o)) = 1$. Hence, for $a > a^*$, a pooling credit with $f = \kappa^L$ is the undefeated equilibrium. If $a \in (0, a^*)$, the undefeated equilibrium is a pooling credit priced at the face value $f \in (\kappa^L, \kappa^H]$ with $\partial f/\partial a \leq 0$, where $f = \kappa^H$ (a pooling asset sale) applies in the limiting case with $a = (0, \tilde{a}]$. In this context it is noteworthy to point out that the separating offer, (q^H, y_S^H, y_C^H, l^H) in (19)-(21), is defeated at all times since the cost of the required asset retention renders the κ^H -consumer's surplus lower than the surplus with a pooling offer with $f = \kappa^L$ for all $a \in (0, a^*]$. Using $u(q) = q^{1-\rho}/(1-\rho)$ with $\rho = 0.5$, Figure 3 visualizes the κ^H -consumer's surplus in a separating equilibrium, $S_c^{sep}(a; \kappa^H)$, and a pooling equilibrium, $S_{c,f=\kappa^L}^{pool}(a; \kappa^H)$ and $S_{c,f=\kappa^H}^{pool}(a; \kappa^H)$, distinguishing between the limiting cases $f = \kappa^L$ and $f = \kappa^H$,

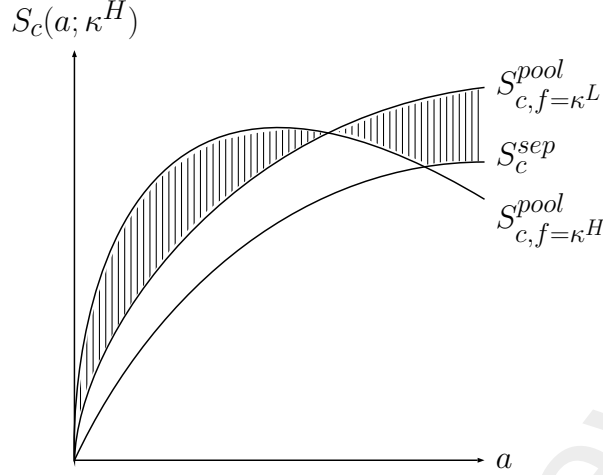


Figure 3: Undefeated equilibrium: high-type consumer surpluses

where $S_{c,f=\kappa^H}^{pool}(a; \kappa^H)$ converges to $S_{c,f=\kappa^L}^{pool}(a; \kappa^H)$ with $f \rightarrow \kappa^L$.

Let us now incorporate the solutions of the bargaining game into the general equilibrium structure of the model to determine the consumer's asset holdings, a , and the equilibrium asset price, ψ , in the CM. Since consumers are homogeneous prior to the realization of the idiosyncratic information shock, their decision does not impart any private information about the assets' future return and each consumer enters the DM with the same amount of assets, a . Proposition D summarizes the general equilibrium.

Proposition D. (Asset pricing) *There exists a unique solution to (11) corresponding to $\psi = \beta R + \mathcal{L}$ with $\mathcal{L} = \partial \mathbb{E} S_c(a; \kappa^x) / \partial a$ representing the liquidity premium, where $\mathcal{L} = 0$ if $A = A^*$ solving $q^* = \beta A^* \kappa^L$, and $\mathcal{L} > 0$ with $\partial \mathcal{L} / \partial A < 0$ if $A < A^*$.*

Proof. See Appendix F.

Proposition D determines the equilibrium asset price, ψ , in the CM and characterizes the relationship between the liquidity premium, $\mathcal{L}$, and the asset supply, A . As shown in Proposition B and C, if assets are costless to carry across periods, i.e., if $\psi = \beta R$, a consumer obtains enough assets to purchase $\bar{q} = q^*$ using a pooling credit, where q^* solves $q^* = \beta A^* \kappa^L$. If $\psi > \beta R$ and $A \in (0, A^*)$, then $\bar{q} < q^*$, and thus $\mathcal{L} > 0$ with the liquidity premium decreasing with an increase in the asset supply, $\partial \mathcal{L} / \partial A < 0$.

4.1 Benchmark Case: Non-recourse Loans

To highlight the effects of the producers' commitment power on equilibrium allocations, this section presents a benchmark scenario in which neither the consumer nor the producer can commit to

future actions (non-recourse loans). In other words, a producer can choose to refuse a consumer's repayment of the credit obligation and keep the pledged collateral instead, should she desire to do so. Vice versa, analog to the main study, a consumer can choose to default on the credit obligation if the repayment exceeds the value of the pledged collateral. Aware of said strategies, a consumer only proposes offers, $\{(\hat{q}^L, \hat{y}_S^L, \hat{y}_C^L, \hat{l}^L), (\hat{q}^H, \hat{y}_S^H, \hat{y}_C^H, \hat{l}^H), (\hat{q}^P, \hat{y}_S^P, \hat{y}_C^P, \hat{l}^P)\}$, that make the producer indifferent between returning the collateral or not, while a producer only accepts offers, i.e., $(\hat{q}, \hat{y}_S, \hat{y}_C, \hat{l}) \in \mathcal{A}_p$, that yield a non-negative surplus in expectation. The consequence is allocation-equivalence of collateralized credit and asset sales despite information frictions. Proposition E presents the undefeated equilibrium.

Proposition E. (Benchmark - undefeated equilibrium) *The undefeated equilibrium satisfying Definition B is characterized by a belief system, $\lambda(\kappa^x \mid (q, y_S, y_C, l))$, and the equilibrium offers, $\{(\hat{q}^L, \hat{y}_S^L, \hat{y}_C^L, \hat{l}^L), (\hat{q}^H, \hat{y}_S^H, \hat{y}_C^H, \hat{l}^H), (\hat{q}^P, \hat{y}_S^P, \hat{y}_C^P, \hat{l}^P)\} \in \mathcal{O}$. For $\kappa^L > 0$, there exists a threshold value $\hat{\pi} \in (0, 1)$ such that:*

- (i) *If $\pi < \hat{\pi}$, then $\hat{S}_c^{pool}(a; \kappa^H) < \hat{S}_c^{sep}(a; \kappa^H)$ and the equilibrium is separating; and*
- (ii) *If $\pi > \hat{\pi}$, then $\hat{S}_c^{pool}(a; \kappa^H) > \hat{S}_c^{sep}(a; \kappa^H)$ and the equilibrium is pooling,*

where $\hat{S}_c^{sep}(a; \kappa^H)$ denotes the high-type consumer's surplus offering a separating offer and $\hat{S}_c^{pool}(a; \kappa^H)$ the high-type consumer's surplus offering a pooling offer. If $\kappa^L = 0$, then a pooling offer is the undefeated equilibrium.

Derivation and Proof: See Appendix G.

Proposition E shows that the high-type consumer's selection between a pooling and a separating offer depends on the share of high-quality assets in the economy, π , and the return of the low-quality asset, κ^L . For $\kappa^L > 0$, if high-quality assets are vast (rare), i.e., if $\pi > \hat{\pi}$ ($\pi < \hat{\pi}$), then a pooling (separating) offer characterizes the undefeated equilibrium. If $\kappa^L = 0$, then the undefeated equilibrium is a pooling offer for any $\pi \in (0, 1)$. Figure 4 visualizes the result distinguishing between $\pi > \hat{\pi}$ in (a) and $\pi < \hat{\pi}$ in (b), where $\hat{C}_0^L$ and $\hat{C}_0^H$ represent the low- and high-type consumers' indifference curves at the lowest possible surplus (i.e., if the producer assigns $\kappa = \kappa^L$), and $\hat{\mathcal{P}}^H$ and $\hat{\mathcal{P}}^P$ represent the producers' participation constraints distinguishing between a separating and a pooling offer, respectively. The undefeated equilibrium is a separating equilibrium if a κ^H -consumer unilaterally has an incentive to deviate to a separating offer, and a pooling equilibrium if both the κ^H -consumer and the κ^L -consumer have an incentive to deviate to a pooling offer.¹⁸

¹⁸The results are in line with Li and Rocheteau (2008) and Bajaj (2018), studying the undefeated equilibrium in a Shi (1995) and Trejos and Wright (1995) second-generation monetary search model with indivisible assets.

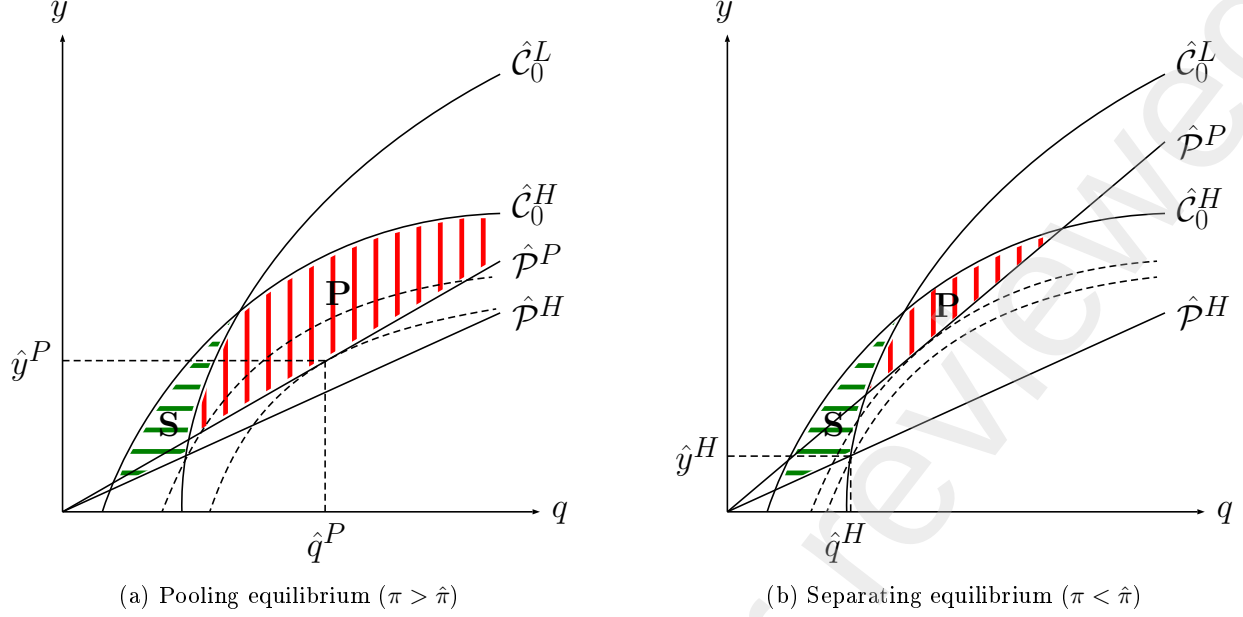


Figure 4: Benchmark case: undefeated equilibrium

5 Money and Assets

The infinite-horizon general equilibrium structure of the model allows for an analysis of long-lived assets alongside the one-period-lived assets discussed to this point. In this section, I alter the environment to study the coexistence of fiat money and information-sensitive short-lived assets to study optimal monetary policy and its effect on the consumers' settlement strategies in the presence of information frictions.¹⁹

The supply of fiat money in the economy is managed by a monetary authority according to the law of motion $M' = (1 + \tau)M$, where M denotes the stock of fiat money in the current period, M' the stock of fiat money in the next period, and expansion/ contraction is conducted through lump-sum transfers, $T = \tau M$, in the CM. One unit of fiat money can buy ϕ units of the numéraire good in the CM. Since the focus lies on symmetric and stationary monetary equilibria, it holds that $M'/M = \phi/\phi' = \gamma$ with γ being the exogenous gross growth rate of the fiat money supply.

An agent $i \in \{c, p\}$ enters the CM with a units of the real asset, m units of fiat money, and liabilities l from the previous period to solve:

$$W_i(a, m, l; \kappa) = \max_{z, h, a', m'} z - h + \mathbb{E}V_i(a', m'; \kappa') \quad (26)$$

$$\text{s.t. } z + \psi a' + \phi m' = h + a\kappa + \phi m + \psi A + T - l, \quad (27)$$

¹⁹Note that the analysis can be reinterpreted to an environment with coexisting liquid information-sensitive and information-insensitive assets, where the latter are long-lived and have a fundamental value of zero. Furthermore, it can be readily extended by exchanging fiat money for long-lived assets with a positive fundamental value and periodical dividend payments.

where the expectation is taken with respect to the assets' stochastic fundamental value, $\kappa^x = \{\kappa^L, \kappa^H\}$. The value functions of a consumer and a producer entering the DM with a units of the real asset and m units of fiat money are:

$$V_c(a, m; \kappa) = u(q) + \beta W(a - y_S, m - d, l; \kappa^x), \quad (28)$$

$$V_p(a, m; \kappa) = -q + \beta W(a + y_S, m + d, l; \kappa^x), \quad (29)$$

where $d \in [0, m]$ denotes the units of fiat money transferred to the producer in exchange for q units of the DM consumption good. In line with Section 4, the bargaining problem has the structure of a signaling game. A strategy for a κ^x -consumer is to specify an offer $(q, y_S, y_C, l, d) \in \mathcal{G} \equiv \mathbb{R}_+ \times [0, a] \times [0, m]$ solving:

$$(q^x, y_S^x, y_C^x, l^x, d^x) \in \arg \max_{q, y_S, y_C, l, d} [u(q) - \beta y_S \kappa^x - \beta y_C f - \beta \phi d] \mathbb{I}_{\mathcal{A}_p} \quad \text{s.t.} \quad (q, y_S, y_C, l, d) \in \mathcal{G}, \quad (30)$$

where the indicator function $\mathbb{I}_{\mathcal{A}_p} = 1$ if $(q, y_S, y_C, l, d) \in \mathcal{A}_p(\lambda)$ with:

$$\mathcal{A}_p(\lambda) = \{-q + \beta y_S \mathbb{E}_\lambda[\kappa_c^x] + \beta l + \beta \phi d \geq 0 \quad \text{and} \quad -l + y_C \mathbb{E}_\lambda[\kappa_c^x] \geq 0\}, \quad (31)$$

and $\mathbb{I}_{\mathcal{A}_p} = 0$ otherwise. In words, a κ^x -consumer maximizes her DM surplus subject to feasibility constraints and the producer's acceptance rule, $\mathcal{A}_p(\lambda)$, where the posterior belief, $\lambda(\kappa^x | (q, y_S, y_C, l, d)) \in [0, 1]$, is determined from the belief system in Appendix A. Proposition F characterizes the undefeated equilibrium.

Proposition F. (Money and assets - undefeated equilibrium) *The undefeated equilibrium satisfying Definition B is characterized by a belief system, $\lambda(\kappa^x | (q, y_S, y_C, l, d))$, and the equilibrium offer, $(q, y_S, y_C, l, d) \in \mathcal{O}$.*

- (i) If $\beta(\phi m + a\kappa^L) \geq q^*$, then $(q, y_S, y_C, l, d) = (\bar{q}, 0, \bar{y}_C, \bar{l}, \bar{d})$ with $\bar{q} = q^*$, $\bar{y}_C = [q^* - \beta\phi\bar{d}]/\beta\kappa^L$, $\bar{l} = \bar{y}_C\kappa^L$, and $\bar{d} = [q^* - \beta\bar{y}_C\kappa^L]/\beta\phi$;
- (ii) If $\beta(\phi m + a[\pi f + (1 - \pi)\kappa^L]) \in (\tilde{q}, q^*)$, then $(q, y_S, y_C, l, d) = (\bar{q}, 0, \bar{y}_C, \bar{l}, \bar{d})$ with $\bar{q} \in (\tilde{q}, q^*)$, $\bar{y}_C = a$, $\bar{l} = \bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ with $f \in (\kappa^L, \kappa^H)$, and $\bar{d} = m$; and
- (iii) If $\beta(\phi m + a[\pi f + (1 - \pi)\kappa^L]) \in (0, \tilde{q}]$, then $(q, y_S, y_C, l, d) = (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}, \bar{d})$, with $\bar{q} \in (0, \tilde{q}]$, $\bar{y}_j = a$, $\bar{l} = \bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ with $f = \kappa^H$, and $\bar{d} = m$.

Derivation and Proof: See Appendix H.

Let us proceed step by step, starting with case (i). If the κ^H -consumer's feasibility constraint is non-binding, consumption is first-best, i.e., $\bar{q} = q^*$, and the transactions are settled using a combination of fiat money and a pooling collateralized credit with the face value $f = \kappa^L$. Since

the pooling credit allows the κ^H -consumer to mimic the κ^L -consumer's complete information offer, the κ^H -consumer has no preference to spend fiat money first, and thus there exists no pecking-order theory of payments despite information frictions. (ii) If the κ^H -consumer's feasibility constraint is binding but the consumer holds enough wealth $a > \tilde{a}$ to purchase $\bar{q} \in (\tilde{q}, q^*)$, where $\partial \tilde{a} / \partial m < 0$, the transaction is settled using a combination of fiat money and a pooling credit with the face value $f \in (\kappa^L, \kappa^H)$. Since the κ^H -consumer is cross-subsidizing the consumption of the κ^L -consumer for all $f \in (\kappa^L, \kappa^H]$, there emerges a pecking-order theory of payment choices where fiat money is the preferred medium of exchange. Lastly, (iii) represents the limiting case with $a \leq \tilde{a}$, and thus $\bar{q} \in (0, \tilde{q}]$, where the κ^H -consumer spends all her fiat money first and settles the remainder by selling assets at the pooling price $R = \pi \kappa^H + (1 - \pi) \kappa^L$, i.e., a collateralized credit with the face value $f = \kappa^H$.

Let us now incorporate the solutions of the bargaining game into the general equilibrium structure of the model to determine the consumers' money and asset holdings, m and a , and the equilibrium prices, ϕ and ψ . Plugging (27) solved for $z - h$ and (28) into (26) yields to consumer's portfolio problem in the CM:

$$(a', m') \in \arg \max_{a', m'} -(\psi - \beta R)a' - (\phi - \beta \phi')m' + \pi S_c(a', m'; \kappa^H) + (1 - \pi)S_c(a', m'; \kappa^L), \quad (32)$$

where $S_c(a, m; \kappa^x)$ denotes the κ^x -consumer's DM surplus, $(\phi - \beta \phi')$ the cost of holding fiat money, and $(\psi - \beta R)$ the cost of holding assets. Hence, a consumer chooses her portfolio depending on the expected benefits fiat money and assets yield in the DM. Furthermore, given the timing of events, the portfolio does not convey any information about the consumer's type, χ .

Definition C. An equilibrium is a list of portfolios, (m, a) , terms of trade in the DM, (q, y_S, y_C, l, d) , and aggregate balances, M and A , such that (m, a) solves (32), (q, y_S, y_C, l, d) solves (30), and the market clearing conditions, $\int_{j \in N} a(j) dj = A$ and $\int_{j \in N} m(j) dj = M$, hold with N being the total number of agents in the economy.

Lemma A. There exists a unique solution to the consumer's portfolio choice, (32):

$$\phi = \beta \phi' + \mathcal{L}_m \quad (33)$$

$$\psi = \beta R + \mathcal{L}_a \quad (34)$$

with the liquidity premia of fiat money and assets:

$$\mathcal{L}_m = \beta \phi' [u'(\bar{q}) - 1], \quad (35)$$

$$\mathcal{L}_a = \beta [\pi f + (1 - \pi) \kappa^L] [u'(\bar{q}) - 1], \quad (36)$$

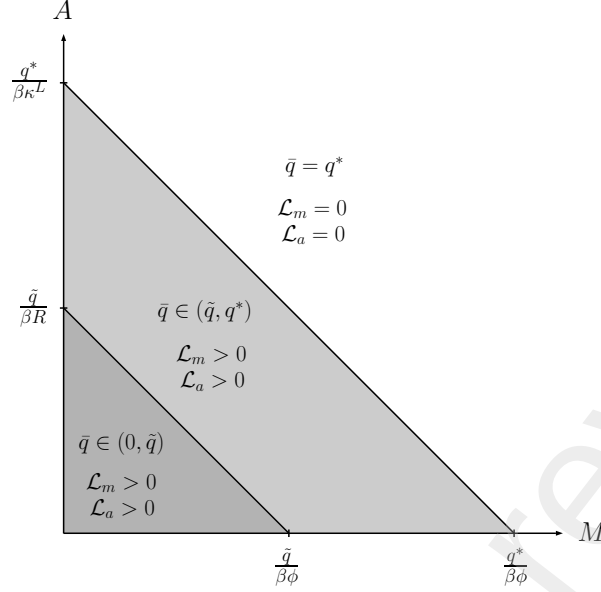


Figure 5: Money and assets: liquidity premia

respectively, where $(\mathcal{L}_m, \mathcal{L}_a) = (0, 0)$ if $\bar{q} = q^*$, and $\mathcal{L}_m > 0$ and $\mathcal{L}_a > 0$ otherwise.

Proof: See Appendix I.

Consider the three cases of the bargaining game: $q = q^*$, $q \in (\tilde{q}, q^*)$, and $q \in (0, \tilde{q}]$. If either fiat money or assets are costless to hold, then a consumer obtains enough liquidity to purchase $q^* = \beta(a\kappa^L + \phi m)$ with $a \in (0, a^*]$ or $m \in (0, m^*]$, and thus $\mathcal{L}_m = \mathcal{L}_a = 0$. If fiat money and assets are costly to hold, i.e., if $\mathcal{L}_m > 0$ and $\mathcal{L}_a > 0$, then $q \in (0, q^*)$. Figure 5 visualizes the results.

Proposition G. (Monetary policy) Assume $A \in (0, A^*)$ where A^* solves $q^* = \beta A^* \kappa^L$. Define the nominal interest rate as $i = \gamma/\beta - 1$, where $\tilde{i}$ corresponds to $\tilde{q} = \beta[AR + \phi m]$. If $i = 0$ (the Friedman rule), then $q = q^*$ for any A . If $0 < i < \tilde{i}$, then $q \in (\tilde{q}, q^*)$ with the equilibrium offer $(q, y_S, y_C, l, d) = (\bar{q}, 0, \bar{y}_C, \bar{l}, \bar{d})$. If $i \geq \tilde{i}$, then $q \in (0, \tilde{q}]$ with the equilibrium offer $(q, y_S, y_C, l, d) = (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}, \bar{d})$.

Proof: See Appendix J.

Proposition G characterizes optimal monetary policy and determines the consumer's optimal payment choice as a function of the nominal interest rate using the general equilibrium results determined in Lemma A. First and foremost, if $A \geq A^*$, then $q = q^*$ for any $i \geq 0$. If $A \in (0, A^*)$, on the other hand, then first-best consumption only occurs at the Friedman rule, i.e., at $i = 0$. Away from the Friedman rule, transactions are settled using a combination of fiat money and a collateralized credit with a face value $f \in (\kappa^L, \kappa^H)$ for moderate levels of inflation, i.e., $0 < i < \tilde{i}$, while selling fiat money and assets at the expected value R is the dominant strategy for $i \geq \tilde{i}$.

Intuitively, if money is fairly inexpensive to hold across periods, a κ^H -consumer holding a units of the information-sensitive asset is better off pledging assets as collateral subject to a haircut than selling them at a discount since the amount of assets needed to support the required over-collateralization is low. With an increase in the cost of holding money, a consumer's money holdings decrease. Once $i \geq \tilde{i}$, then $a \leq \tilde{a}$, and thus $S_{c,f=\kappa^H}^{pool}(a, m; \kappa^H) > S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a, m; \kappa^H)$ despite the cross-subsidization of the κ^L -consumer.

6 Conclusion

This paper studies optimal payment strategies in over-the-counter markets subject to commitment and information frictions. Agents, privately informed about the quality of the assets in their portfolio, choose whether to settle transactions instantaneously by outright selling assets or using the assets as collateral to secure a bilateral credit agreement. Bargaining under private information determines the terms of trade, where the refinement of the undefeated equilibrium endogenously selects between separating and pooling offers.

The results show that collateralized credit and asset sales can be non-equivalent if the lender can commit to returning the pledged collateral upon the borrower's repayment of the agreed-upon credit obligation. If the aggregate asset supply is large, agents prefer a pooling collateralized credit subject to a haircut over any other sale or collateralized credit agreement. If assets are scarce, on the other hand, the optimal payment strategy is to sell assets outright at the expected market value.

An extension incorporating co-existing information-sensitive assets and fiat money further sheds new light on portfolio management under adverse selection and ties optimal payment strategies to monetary policy. The results show that conditional on the supply of information-sensitive assets in the economy, first-best allocations can be obtained for a positive range of nominal interest rates. Once information-sensitive assets are scarce, cross-subsidization characterizes the terms of trade, rendering the Friedman rule the unique optimal monetary policy.

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Appendix: Definitions and Proofs

A. Definition of the Belief System

To formally define the belief system used to identify the separating and pooling offers, it is helpful to first identify the lowest possible surplus, $S_{c,0}(a; \kappa^x)$, a κ^x -consumer can obtain in equilibrium. It is given by:

$$S_{c,0}(a; \kappa^x) = \max_{q, y_S, y_C, l} u(q) - \beta y_S \kappa^x - \beta l, \quad (\text{A.1})$$

$$\text{s.t. } -q + \beta y_S \kappa^L + \beta l \geq 0, \quad (\text{A.2})$$

$$\text{s.t. } -l + y_C \kappa^L \geq 0, \quad (\text{A.3})$$

$$\text{s.t. } -y_S - y_C + a \geq 0. \quad (\text{A.4})$$

Next, let $S_c(a; \kappa^x) = u(q) - \beta y_S \kappa^x - \beta y_C f$ be a κ^x -consumer's surplus for some $(q, y_S, y_C, l) \in \mathcal{O}$ where $\mathcal{O}$ is the set of equilibrium offers. The belief system $\lambda(\kappa^x \mid (q, y_S, y_C, l))$ is given by:

$$\lambda(\kappa^H \mid (q, y_S, y_C, l)) = \frac{\pi \mathbb{I}[S_c(a; \kappa^H) > S_{c,0}(a; \kappa^H)]}{\pi \mathbb{I}[S_c(a; \kappa^H) > S_{c,0}(a; \kappa^H)] + (1 - \pi) \mathbb{I}[S_c(a; \kappa^L) > S_{c,0}(a; \kappa^L)]}, \quad (\text{A.5})$$

$$\lambda(\kappa^L \mid (q, y_S, y_C, l)) = \frac{(1 - \pi) \mathbb{I}[S_c(a; \kappa^L) \geq S_{c,0}(a; \kappa^L)]}{(1 - \pi) \mathbb{I}[S_c(a; \kappa^L) \geq S_{c,0}(a; \kappa^L)] + \pi \mathbb{I}[S_c(a; \kappa^H) > S_{c,0}(a; \kappa^H)]}, \quad (\text{A.6})$$

where the indicator function $\mathbb{I} = 1$ if the statement in the brackets is true, and else $\mathbb{I} = 0$. Any out-of equilibrium offer, $(q^o, y_S^o, y_C^o, l^o) \notin \mathcal{O}$, is attributed to a low-type consumer, and thus $\lambda(\kappa^L \mid (q^o, y_S^o, y_C^o, l^o)) = 1$.

B. Definition of Undeclared Equilibria

Borrowing from [Mailath et al. \(1993\)](#) and [Wang \(2020\)](#), I provide a formal definition of undeclared equilibria for an environment with two or more types of consumers, where $\chi = 1, 2, \dots, X$ denotes the types of consumers and π_χ the corresponding proportions of type χ consumers in the environment.

Definition B.1. A PBE characterized by $\{(q^\chi, y_S^\chi, y_C^\chi, l^\chi)\}_{\chi=1}^X$ and $\lambda(\chi \mid (q, y_S, y_C, l))$ is defeated by an alternative PBE characterized by $\{(q'^\chi, y_S'^\chi, y_C'^\chi, l'^\chi)\}_{\chi=1}^X$ and $\lambda(\chi \mid (q', y_S', y_C', l'))$ if:

(i) There exists an offer $(q'', y_S'', y_C'', l'') \in \{(q'^\chi, y_S'^\chi, y_C'^\chi, l'^\chi)\}_{\chi=1}^X$ such that $(q'', y_S'', y_C'', l'') \notin \{(q^\chi, y_S^\chi, y_C^\chi, l^\chi)\}_{\chi=1}^X$, and $\Omega = \{\chi \mid (q'^\chi, y_S'^\chi, y_C'^\chi, l'^\chi) = (q'', y_S'', y_C'', l'')\} \neq \emptyset$;

(ii) For all $\chi \in \Omega$, the consumers' utility is (weakly) higher in the alternative PBE, while for some

$\chi \in \Omega$ the consumers' utility is strictly higher in the alternative PBE; and

(iii) There exist $\chi' \in \Omega$ such that $\lambda(\chi' \mid (q'', y_S'', y_C'', l'')) \neq \pi_{\chi'} \theta(\chi') / (\sum_{\chi=1}^X \pi_{\chi} \theta(\chi))$ for any $\theta : \{L, H\} \rightarrow [0, 1]$ satisfying:

(a) $\theta(\chi) = 1$ if $\chi \in \Omega$ and a strictly higher utility in the alternative PBE; and

(b) $\theta(\chi) = 0$ if $\chi \notin \Omega$.

C. Proof of Proposition A

The proof proceeds in three steps. First, in part (a), I determine the least-inefficient separating offers for the low-type and the high-type consumer, respectively. Second, in part (b), I show that the proposed offers satisfy Definition B. Third, in part (c), I determine the comparative statics.

Part (a): Let us start with the low-type consumer. Given the system of beliefs defined in Appendix A, a producer only attributes an equilibrium offer $(q^L, y_S^L, y_C^L, l^L) \in \mathcal{O}$ to a low-type consumer, i.e., $\lambda(\kappa^L \mid (q^L, y_S^L, y_C^L, l^L)) = 1$, if the offer makes the low-type weakly better off than $S_{c,0}(a; \kappa^L)$, while making the high-type weakly worse off than $S_{c,0}(a; \kappa^H)$. Since the offer in (A.1)-(A.4) identifying the lowest possible surplus is the only equilibrium offer satisfying this requirement, a low-type consumer has no choice but to reveal her type and make a complete-information offer yielding the equilibrium allocations in (16)-(18) if $\kappa^L > 0$, and $q^L = y_j^L = l^L = 0$ if $\kappa^L = 0$.

Let us now consider the high-type consumer. In order for a producer to believe $\lambda(\kappa^H \mid (q^H, y_S^H, y_C^H, l^H)) = 1$, the offer $(q^H, y_S^H, y_C^H, l^H) \in \mathcal{O}$ needs to make the high-type consumer strictly better off than the lowest possible surplus, $S_{c,0}(a; \kappa^H)$, while making the low-type consumer weakly worse off than $S_{c,0}(a; \kappa^L)$. Thus, a high-type consumer solves:

$$(q^H, y_S^H, y_C^H, l^H) \in \arg \max_{q, y_S, y_C, l} u(q) - \beta y_S \kappa^H - \beta l, \quad (\text{C.1})$$

$$\text{s.t. } S_{c,0}(a; \kappa^L) \geq u(q) - \beta y_S \kappa^L - \beta y_C \kappa^L, \quad (\text{C.2})$$

$$\text{s.t. } -q + \beta y_S \kappa^H + \beta l \geq 0, \quad (\text{C.3})$$

$$\text{s.t. } -l + y_C \kappa^H \geq 0, \quad (\text{C.4})$$

$$\text{s.t. } -y_S - y_C + a \geq 0, \quad (\text{C.5})$$

where (C.2) corresponds to an incentive-compatibility constraint ensuring that the low-type consumer has no incentive to mimic the offer of the high-type consumer. To obtain (19)-(21), set the incentive compatibility constraint, (C.2), to equality and plug in the producer's participation and the collateral constraints, (C.3) and (C.4). The allocations imply $q^H < q^L \leq q^*$ and $y^H < y^L \leq a$, i.e., high-type consumers consume less than low-type consumers and retain a fraction of their high-return assets. The fact that $q^L = q^H = y_j^L = y_j^H = 0$ if $\kappa^L = 0$ is straightforward from (A.1)-(A.4)

and (C.2).

Part (b): As shown above, the proposed separating offers, $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H)\}$, satisfy conditions (i) and (ii) in Definition B. Further, among all separating equilibria, the proposed equilibrium maximizes the utility of the high-type consumer, and thus the high-type consumer has no incentive to deviate. Consequentially, the proposed equilibrium is undefeated, satisfying condition (iii) in Definition B.

Part (c): Let us now determine the comparative statics for q^H and y^H if $\kappa^L > 0$. Given (19), for $a > 0$, q^H is the unique solution in $[0, q^L]$ to:

$$u(q^L) - q^L = u(q^H) - q^H \frac{\kappa^L}{\kappa^H}. \quad (\text{C.6})$$

To determine the signs of $\partial q^H / \partial(\kappa^H - \kappa^L)$ and $\partial y^H / \partial(\kappa^H - \kappa^L)$, differentiate (C.6) to obtain:

$$\frac{\partial q^H}{\partial \kappa^L} = \frac{[u'(q^L) - 1]\beta a + \beta y^H}{u'(q^H) - \frac{\kappa^L}{\kappa^H}} > 0, \quad (\text{C.7})$$

$$\frac{\partial q^H}{\partial \kappa^H} = -\frac{\frac{\kappa^L}{\kappa^H} \beta y^H}{u'(q^H) - \frac{\kappa^L}{\kappa^H}} < 0, \quad (\text{C.8})$$

suggesting $\partial q^H / \partial(\kappa^H - \kappa^L) < 0$ for $q^L \in (0, q^*]$, where $q^L = \beta y^L \kappa^L$ with $y^L \in (0, a]$, $q^H = \beta y^H \kappa^H$, and $0 < y^H < y^L$. Using $q^H = \beta y^H \kappa^H$, differentiating, and combining with $\frac{\partial q^H}{\partial \kappa^H}$ and $\frac{\partial q^H}{\partial \kappa^L}$ in (C.7) and (C.8), yields:

$$\frac{\partial y^H}{\partial \kappa^L} = \frac{[u'(q^L) - 1]a + y^H}{\kappa^H u'(q^H) - \kappa^L} > 0, \quad (\text{C.9})$$

$$\frac{\partial y^H}{\partial \kappa^H} = -\frac{u'(q^H)y^H}{\kappa^H u'(q^H) - \kappa^L} < 0, \quad (\text{C.10})$$

suggesting $\partial y^H / \partial(\kappa^H - \kappa^L) < 0$ for $q^L \in (0, q^*]$. To determine the sign of $\partial q^H / \partial a$, differentiate (C.6) and use $\frac{\partial q^L}{\partial a} = \beta \kappa^L$ to obtain:

$$\frac{\partial q^H}{\partial a} = \frac{\beta \kappa^L [u'(q^L) - 1]}{u'(q^H) - \frac{\kappa^L}{\kappa^H}}, \quad (\text{C.11})$$

where $\partial q^H / \partial a > 0$ if $q^L < q^*$, and $\partial q^H / \partial a = 0$ if $q^L = q^*$. To determine the sign of $\partial y^H / \partial a$ use $q^H = \beta y^H \kappa^H$ and differentiate to obtain:

$$\frac{\partial q^H}{\partial a} = \beta \kappa^H \frac{\partial y^H}{\partial a}. \quad (\text{C.12})$$

Combine (C.11) and (C.12) to obtain:

$$\frac{\partial y^H}{\partial a} = \frac{u'(q^L) - 1}{\frac{\kappa^H}{\kappa^L} u'(q^H) - 1}, \quad (\text{C.13})$$

where $\partial y^H / \partial a \in (0, 1)$ for $q^L < q^*$, since $u'(q^L) < u'(q^H)$ and $\kappa^L < \kappa^H$. For $q^L = q^*$, and hence $u'(q^L) = 1$, however, $\partial y^H / \partial a = 0$.

Lastly, to determine the comparative statics for q^L and y^L , use the equilibrium condition $u'(q^L) - 1 \geq 0$ with $q^L = \beta y^L \kappa^L$. Taking the derivative with respect to κ^L and a , respectively, yields $\partial q^L / \partial \kappa^L > 0$, $\partial y_j^L / \partial \kappa^L = 0$, $\partial q^L / \partial a > 0$, and $\partial y_j^L / \partial a = 1$ if $q^L < q^*$, and $\partial q^L / \partial \kappa^L = 0$, $\partial y_j^L / \partial \kappa^L < 0$, and $\partial q^L / \partial a = \partial y_j^L / \partial a = 0$ if $q^L = q^*$. ■

D. Proof of Proposition B

The proof proceeds in three steps. First, in part (a), I determine the most-efficient pooling offer. Second, in part (b), I show that the proposed offer satisfies Definition B. Third, in part (c), I determine the comparative statics.

Part (a): Given the belief system defined in Appendix A, a pooling offer, $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \mathcal{O}$, does not reveal the consumer's type, i.e., $\lambda(\kappa^H \mid (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})) = \pi$ and $\lambda(\kappa^L \mid (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})) = 1 - \pi$ if $S_c(a; \kappa^L) > S_{c,0}(a; \kappa^L)$ and $S_c(a; \kappa^H) > S_{c,0}(a; \kappa^H)$ holds. Consequently, given the producer's acceptance rule in (14), a producer accepts the proposed offer if her expected profit is non-negative, i.e., if $-\bar{q} + \beta \bar{y}_S [\pi \kappa^H + (1 - \pi) \kappa^L] + \beta \bar{y}_C [\pi f + (1 - \pi) \kappa^L] \geq 0$, where $f \in [\kappa^L, \kappa^H]$ denotes the face value of the loan. Thus, a high-type consumer solves:

$$(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}) \in \arg \max_{q, y_S, y_C, l} u(q) - \beta y_S \kappa^H - \beta y_C f, \quad (\text{D.1})$$

$$\text{s.t. } u(q) - \beta y_S \kappa^L - \beta y_C \kappa^L \geq S_{c,0}(a; \kappa^L), \quad (\text{D.2})$$

$$\text{s.t. } -q + \beta y_S [\pi \kappa^H + (1 - \pi) \kappa^L] + \beta l \geq 0, \quad (\text{D.3})$$

$$\text{s.t. } -l + y_C [\pi f + (1 - \pi) \kappa^L] \geq 0, \quad (\text{D.4})$$

$$\text{s.t. } -y_S - y_C + a \geq 0. \quad (\text{D.5})$$

Starting with an asset sale assuming $y_C = l = 0$, solving (D.3) (holding with equality) for $\bar{y}_S = \bar{q} / \beta R$, plugging into the high-type consumer's maximization problem, (D.1), and taking the derivative with respect to $\bar{q}$ yields:

$$u'(\bar{q}) - \kappa^H / [\pi \kappa^H + (1 - \pi) \kappa^L] \geq 0, \quad (\text{D.6})$$

with $\bar{y}_S \in (0, a]$, where $\bar{q} < q^*$ solves (D.6) holding with equality. Vice versa, considering a pooling credit assuming $\bar{y}_S = 0$, solving (D.3) and (D.4) combined (holding with equality) for

$\bar{y}_C = \bar{q}/\beta[\pi f + (1 - \pi)\kappa^L]$, plugging into (D.1), and taking the derivative with respect to $\bar{q}$ yields:

$$u'(\bar{q}) - f/[\pi f + (1 - \pi)\kappa^L] \geq 0, \quad (D.7)$$

with $\bar{y}_C \in (0, a]$, $\bar{q} \in (0, q^*]$, and $f \in [\kappa^L, \kappa^H]$, where $\bar{q} = q^*$ if $f = \kappa^L$ and $a \geq a^*$. Having determined the distinction between a sale and a credit in the pooling offer, what remains to be determined are the threshold asset holdings, $\tilde{a} \in (0, a^*)$, above which a κ^H -consumer prefers a pooling credit with $f \in [\kappa^L, \kappa^H]$ and below which a pooling credit with $f = \kappa^H$ (i.e., a pooling asset sale) is the preferred option. First and foremost, for $a \geq a^*$, given the allocation $\bar{q} = \beta\bar{y}_C[\pi f + (1 - \pi)\kappa^L] \in (0, q^*]$ in (22), it holds that the high-type consumer's surplus $S_{c,f=\kappa^L}^{pool}(q^*) > S_{c,f=\kappa^H}^{pool}(\tilde{q}_S)$ where $\tilde{q}_S$ solves $u'(\tilde{q}_S) = \kappa^H/[\pi\kappa^H + (1 - \pi)\kappa^L]$, and thus a pooling credit with $f = \kappa^L$ dominates a credit with $f = \kappa^H$. Second, given the concavity of $u(\cdot)$, the comparative statics show that both surpluses – distinguishing between $f \in [\kappa^L, \kappa^H]$ and $f = \kappa^H$ – are hump-shaped, satisfying:

$$\frac{\partial S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a; \kappa^H)}{\partial a} = \begin{cases} u'(\bar{q}) - \frac{f}{\pi f + (1 - \pi)\kappa^L} < 0 & \text{if } \bar{q} > \tilde{q}_C \\ u'(\bar{q}) - \frac{f}{\pi f + (1 - \pi)\kappa^L} = 0 & \text{if } \bar{q} = \tilde{q}_C \\ u'(\bar{q}) - \frac{f}{\pi f + (1 - \pi)\kappa^L} > 0 & \text{if } \bar{q} < \tilde{q}_C, \end{cases} \quad (D.8)$$

and:

$$\frac{\partial S_{c,f=\kappa^H}^{pool}(a; \kappa^H)}{\partial a} = \begin{cases} u'(\bar{q}) - \frac{\kappa^H}{\pi\kappa^H + (1 - \pi)\kappa^L} < 0 & \text{if } \bar{q} > \tilde{q}_S \\ u'(\bar{q}) - \frac{\kappa^H}{\pi\kappa^H + (1 - \pi)\kappa^L} = 0 & \text{if } \bar{q} = \tilde{q}_S \\ u'(\bar{q}) - \frac{\kappa^H}{\pi\kappa^H + (1 - \pi)\kappa^L} > 0 & \text{if } \bar{q} < \tilde{q}_S, \end{cases} \quad (D.9)$$

with $\partial S_{c,f=\kappa^H}^{pool}(a; \kappa^H)/\partial a|_{\lim_{a \rightarrow 0}} - \partial S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a; \kappa^H)/\partial a|_{\lim_{a \rightarrow 0}} > 0$, implying a single-crossing of $S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a; \kappa^H)$ and $S_{c,f=\kappa^H}^{pool}(a; \kappa^H)$ given $\tilde{q}_S < q^*$. Consequentially, there exists a threshold asset holding, $\tilde{a} \in (0, a^*)$, solving:

$$S_{c,f \in [\kappa^L, \kappa^H]}^{pool} \equiv u(\beta\tilde{a}[\pi f + (1 - \pi)\kappa^L]) - \beta\tilde{a}f = u(\tilde{q}_S) - \tilde{q}_S\kappa^H/R \equiv S_{c,f=\kappa^H}^{pool}, \quad (D.10)$$

with:

$$S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a; \kappa^H) - S_{c,f=\kappa^H}^{pool}(a; \kappa^H) \begin{cases} > 0 & \text{if } a > \tilde{a} \\ = 0 & \text{if } a = \tilde{a} \\ < 0 & \text{if } a < \tilde{a}. \end{cases} \quad (D.11)$$

Thus, for $a > \tilde{a}$, a pooling collateralized credit with $f \in [\kappa^L, \kappa^H]$ dominates an asset sale, while for $a < \tilde{a}$, a pooling credit with $f = \kappa^H$ (i.e., an asset sale) is the dominant strategy.

Part (b): As shown above, the proposed pooling offer, $(\bar{q}, \bar{y}_S, \bar{y}_C, l)$, satisfies conditions (i) and (ii) in Definition B. Further, among all pooling equilibria, the proposed offer maximizes the

utility of the high-type consumer, and thus the high-type consumer has no incentive to deviate to a different pooling offer. Consequentially, the proposed offer is unique and undefeated, satisfying condition (iii) in Definition B.

Part (c): Lastly, let us determine the comparative statics. Differentiating the equilibrium condition (D.7) with respect to f , κ^L , and π using $\bar{q} = \beta \bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ yields:

$$\frac{\partial \bar{q}}{\partial f} = \frac{(1 - \pi)\kappa^L}{u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^2} < 0, \quad (\text{D.12})$$

$$\frac{\partial \bar{q}}{\partial \kappa^L} = -\frac{(1 - \pi)\kappa^H}{u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^2} > 0, \quad (\text{D.13})$$

$$\frac{\partial \bar{q}}{\partial \pi} = -\frac{f(f - \kappa^L)}{u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^2} \geq 0, \quad (\text{D.14})$$

where (D.14) holds with equality if $f = \kappa^L$, and:

$$\frac{\partial \bar{y}_C}{\partial f} = \frac{(1 - \pi)\kappa^L - \pi u''(\bar{q})\bar{q}[\pi f + (1 - \pi)\kappa^L]}{\beta u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^3} < 0, \quad (\text{D.15})$$

$$\frac{\partial \bar{y}_C}{\partial \kappa^L} = -\frac{(1 - \pi)[f + u''(\bar{q})\bar{q}[\pi f + (1 - \pi)\kappa^L]]}{\beta u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^3} \geq 0, \quad (\text{D.16})$$

$$\frac{\partial \bar{y}_C}{\partial \pi} = -\frac{(f - \kappa^L)[f + u''(\bar{q})\bar{q}[\pi f + (1 - \pi)\kappa^L]]}{\beta u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]^3} \geq 0, \quad (\text{D.17})$$

with $\partial \bar{y}_C / \partial \kappa^L > 0$ and $\partial \bar{y}_C / \partial \pi > 0$ if $f > |u''(\bar{q})\bar{q}[\pi f + (1 - \pi)\kappa^L]|$ and $f > \kappa^L$, and negative otherwise. Lastly, differentiating (D.7) and $\bar{q} = \beta a[\pi f + (1 - \pi)\kappa^L]$ with respect to a yields:

$$u''(\bar{q})\frac{\partial \bar{q}}{\partial a} = \frac{\partial f}{\partial a} \frac{(1 - \pi)\kappa^L}{[\pi f + (1 - \pi)\kappa^L]^2}, \quad (\text{D.18})$$

and:

$$\frac{\partial \bar{q}}{\partial a} = \beta[\pi f + (1 - \pi)\kappa^L] + \beta a \pi \frac{\partial f}{\partial a}, \quad (\text{D.19})$$

respectively. Combining gives:

$$\frac{\partial f}{\partial a} = \frac{\beta u''(\bar{q})[\pi f + (1 - \pi)\kappa^L]}{\frac{(1 - \pi)\kappa^L}{[\pi f + (1 - \pi)\kappa^L]^2} - u''(\bar{q})\beta a \pi} < 0. \quad (\text{D.20})$$

In words, the face value of the loan decreases with the consumer's asset holdings with $f \in (\kappa^L, \kappa^H)$ for $a \in (\tilde{a}, a^*)$ and the limits $f = \kappa^L$ if $a > a^*$ and $f = \kappa^H$ if $a < \tilde{a} \in (0, a^*)$. ■

E. Proof of Proposition C

To determine the undefeated equilibrium, i.e., the choice between a pooling and a separating offer,

the proof proceeds in two steps. First, in step (a), to satisfy condition (iii) in Definition B, I show that an equilibrium is undefeated if it maximizes the high-type consumer's surplus, $S_c(a; \kappa^H)$, and if there exists no other equilibrium offer such that either the high-type consumer has an incentive to unilaterally deviate to a separating offer (i.e., $\Omega = H$), or both types prefer a deviation to a pooling offer (i.e., $\Omega = \{L, H\}$). In a second step (b), I show that the separating equilibrium is defeated at all times.

Part (a): Recall from Proposition A and Proposition B, the κ^H -consumer's surpluses for a separating and a pooling offer are:

$$S_c^{pool}(a; \kappa^H) = u(\bar{q}) - \bar{q}f/[\pi f + (1 - \pi)\kappa^L], \quad (\text{E.1})$$

$$S_c^{sep}(a; \kappa^H) = u(q^H) - q^H, \quad (\text{E.2})$$

respectively, with $\bar{q} = \beta \bar{y}_C[\pi f + (1 - \pi)\kappa^L] \in (0, q^*]$ for $f \in [\kappa^L, \kappa^H]$ and $q^H = \beta y_j^H \kappa^H \in (0, q^*)$ for $j \in \{\mathcal{S}, \mathcal{C}\}$ defined in (19) and (22). If the pooling offer yields a higher surplus for the high-type consumer than the separating equilibrium, a high-type consumer has an incentive to deviate to a pooling offer. Furthermore, a low-type consumer is weakly-better off in the pooling equilibrium, and thus both types have an incentive to deviate. To see this, let us compare $S_c^{pool}(a; \kappa^H)$ and $S_c^{sep}(a; \kappa^H)$ distinguishing between $f \in (\kappa^L, \kappa^H]$ and $f = \kappa^L$. For $f \in (\kappa^L, \kappa^H]$, a high type consumer is better off offering a pooling offer if:

$$u(\bar{q}) - \bar{q} \frac{f}{\pi f + (1 - \pi)\kappa^L} > u(q^H) - q^H, \quad (\text{E.3})$$

and thus if $\bar{q} > q^H$ since $f/[\pi f + (1 - \pi)\kappa^L] > 1$ for $f > \kappa^L$. To show the effect on a low-type consumer's surplus, extend (E.3) yielding:

$$u(\bar{q}) - \bar{q} \frac{\kappa^L}{\pi f + (1 - \pi)\kappa^L} + \bar{q} \left(\frac{\kappa^L - f}{\pi f + (1 - \pi)\kappa^L} \right) > u(q^H) - q^H \frac{\kappa^L}{\kappa^H} + q^H \left(\frac{\kappa^L - \kappa^H}{\kappa^H} \right), \quad (\text{E.4})$$

and thus (using the binding incentive-compatibility constraint $u(q^L) - q^L = u(q^H) - q^H \frac{\kappa^L}{\kappa^H}$ in (19)):

$$u(\bar{q}) - \bar{q} \frac{\kappa^L}{\pi f + (1 - \pi)\kappa^L} > u(q^L) - q^L, \quad (\text{E.5})$$

since $\bar{q} > q^H$ and $\bar{q} \left(\frac{\kappa^L - f}{\pi f + (1 - \pi)\kappa^L} \right) < q^H \left(\frac{\kappa^L - \kappa^H}{\kappa^H} \right) < 0$, where the left hand side of (E.5) corresponds to a low-type consumer's surplus in the pooling offer and the right hand side of (E.5) to a low-type consumer's surplus in the separating offer. Hence, the separating offer is defeated by a pooling offer if both the low- and the high-type consumer have a preference to deviate. Vice versa, it is straightforward to show that the pooling offer is defeated by the separating offer if it offers the high-type consumer a strictly higher surplus, i.e., if high-type consumers have an incentive to

deviate unilaterally. Second, let us compare $S_c^{pool}(a; \kappa^H)$ and $S_c^{sep}(a; \kappa^H)$ for $f = \kappa^L$. A high type consumer is better off offering a pooling offer if:

$$u(\bar{q}) - \bar{q} > u(q^H) - q^H. \quad (\text{E.6})$$

Since $q^H \in [0, q^L)$ in (19) and $\bar{q} = \min\{\beta a \kappa^L, q^*\}$ for $f = \kappa^L$, it is apparent that a pooling offer defeats the separating offer for $f = \kappa^L$, since it offers a strictly higher surplus to the high-type consumer, while a low-type consumer is equally well off than in the separating offer.

Part (b): This part follows directly from part (a). Since a pooling credit with $f = \kappa^L$ defeats a separating offer for any $a \in (0, a^*]$, i.e., $S_{c, f=\kappa^L}^{pool}(a; \kappa^H) - S_c^{sep}(a; \kappa^H) > 0 \forall a \in (0, a^*]$, the separating offer is defeated at all times. ■

F. Proof of Proposition D

The equilibrium asset price, ψ , solves (11) altered to incorporate the stochastic asset returns, $\kappa = \{\kappa^L, \kappa^H\}$:

$$a \in \arg \max_a -(\psi - \beta R)a + \mathbb{E}S_c(a; \kappa^x) \geq 0, \quad (\text{F.1})$$

where $\mathbb{E}S_c(a; \kappa^x) = \pi S_c(a; \kappa^H) + (1 - \pi)S_c(a; \kappa^L)$ with $S_c(a; \kappa^x)$ denoting the surplus of a κ^x -consumer in the DM. Definition F.1 defines the equilibrium.

Definition F.1. *An equilibrium in the CM is a list of asset holdings, a , terms of trade in the DM, $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H), (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})\} \in \mathcal{O}$, and the aggregate asset supply, A , such that a solves (F.1), $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H), (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})\} \in \mathcal{O}$ solves (15), and the market clearing condition, $\int_{j \in N} a(j) dj = A$, holds.*

The first order condition of the agents' CM problem (F.1) is:

$$\psi = \beta R + \mathcal{L}, \quad (\text{F.2})$$

with:

$$\mathcal{L}^{sep} = \beta(1 - \pi)\kappa^L[u'(q^L) - 1] + \beta\pi\kappa^H[u'(q^H) - 1] \frac{[u'(q^L) - 1]}{[(\kappa^H/\kappa^L)u'(q^H) - 1]}, \quad (\text{F.3})$$

$$\mathcal{L}^{pool} = \beta[\pi f + (1 - \pi)\kappa^L][u'(\bar{q}) - 1], \quad (\text{F.4})$$

and $\{q^L, q^H, \bar{q}\}$ defined in (16), (19), and (22), where $\mathcal{L}^{sep}$ denotes the liquidity premium in a separating equilibrium and $\mathcal{L}^{pool}$ the liquidity premium in a pooling equilibrium. If $a = a^*$, then $\bar{q} = q^*$ with $f = \kappa^L$ suggesting $\mathcal{L}^{pool} = 0$, and thus $\psi = \beta R$. If $a \in (0, a^*)$, then $\bar{q} \in (0, q^*)$, and thus $\mathcal{L}^{pool} > 0$ with $\partial \mathcal{L}^{pool} / \partial A < 0$. ■

G. Proof of Proposition E

The proof starts with two auxiliary Lemmata, G.1 and G.2, characterizing the separating and pooling offers in the absence of the producer's ability to commit, respectively.

Lemma G.1. *In the absence of commitment, the unique separating equilibrium $\{(\hat{q}^L, \hat{y}_S^L, \hat{y}_C^L, \hat{l}^L), (\hat{q}^H, \hat{y}_S^H, \hat{y}_C^H, \hat{l}^H)\} \in \mathcal{O}$ satisfying Definition B is equivalent to $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H)\} \in \mathcal{O}$ in (16)-(18) and (19)-(21).*

Proof of Lemma G.1. As shown in Proposition A, the separating offers (16)-(18) and (19)-(21) eliminate the producer's uncertainty regarding the future return of the provided asset and allow for future credit obligations to be priced at the asset's fundamental value, κ^X . Consequentially, there is neither an incentive for the consumer to default nor for the producer to renege on her promise to return the collateral upon repayment of the credit obligation, suggesting allocation equivalence in the absence of a producer's ability to commit. ■

Lemma G.2. *In the absence of commitment, the unique pooling equilibrium $(\hat{q}^P, \hat{y}_S^P, \hat{y}_C^P, \hat{l}^P) \in \mathcal{O}$ satisfying Definition B is equivalent to $(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})$ in (22)-(24) with $f = \kappa^H$, and thus $\hat{q}^P = \beta \hat{y}_j^P [\pi \kappa^H + (1 - \pi) \kappa^L] \in (0, q^*)$, $\hat{y}_j^P = \hat{q}^P / \beta [\pi \kappa^H + (1 - \pi) \kappa^L] \in (0, a]$, and $\hat{l}^P = \hat{y}_C^P R$.*

Proof of Lemma G.2. Proof by contradiction. Assume assets are offered as collateral at a face value $f < \kappa^H$ with $\hat{q}^P = \beta \hat{y}_C^P [\pi f + (1 - \pi) \kappa^L]$ and $\hat{l}^P = \hat{y}_C^P [\pi f + (1 - \pi) \kappa^L]$ analog to (22)-(24) in Proposition B. In said scenario, a producer has a strict incentive to renege on the promise to return the collateral since defaulting yields a positive expected surplus, i.e.,:

$$\mathbb{E}S_p^D = -\hat{q}^P + \beta \hat{y}_C^P [\pi \kappa^H + (1 - \pi) \kappa^L] = \beta \hat{y}_C^P \pi (\kappa^H - f) > 0, \quad (\text{G.1})$$

for any $f < \kappa^H$. For a high-type consumer, such an offer would be equivalent to selling the high-return assets at the price $f < \kappa^H$, resulting in the consumer's surplus:

$$S_c(a; \kappa^H) = u(\hat{q}^P) - \beta \hat{y}_C^P \kappa^H. \quad (\text{G.2})$$

Since such an offer yields a positive surplus for the producer, it is strictly dominated by selling assets at the pooling price, R , yielding an expected surplus of zero to the producer. Consequently, high-type consumers only pledge their assets as collateral if priced at the face value $f = \kappa^H$, rendering sale and collateralized credit equivalent. ■

Having characterized the separating and pooling offers in the absence of the producer's ability to commit, we now determine the undefeated equilibrium. From (E.1) and (E.2) we know that a high-type consumer's surplus in a separating equilibrium is independent of π , and monotonically increasing in π in a pooling equilibrium. Consider two limiting cases: If $\pi \rightarrow 1$, since a high-type consumer's surplus is independent of π in a separating equilibrium, consumption $\hat{q}^H \in [0, \hat{q}^L)$ is equivalent to (19) in Proposition A. In a pooling equilibrium, since $R \rightarrow \kappa^H$ with $\pi \rightarrow 1$, it follows that $\hat{q}^P \rightarrow q^*$ given $\max_{\hat{q}^P} u(\hat{q}^P) - \hat{q}^P \frac{\kappa^H}{R}$. Hence, for $\pi \rightarrow 1$, it follows that $\hat{S}_c^{pool}(a; \kappa^H) > \hat{S}_c^{sep}(a; \kappa^H)$. If $\pi \rightarrow 0$, on the other hand, then $R \rightarrow \kappa^L$, and hence the high-type consumer's problem offering a pooling sale reduces to $\max_{\hat{q}^P} u(\hat{q}^P) - \hat{q}^P \frac{\kappa^H}{\kappa^L}$, suggesting $\hat{q}^P < q^L$. Consequentially, the incentive compatibility constraint, (19), holds at $(\hat{q}^P, \hat{y}_j^P)$, i.e., $u(\hat{q}^P) - \beta \hat{y}_j^P \kappa^L \leq u(\hat{q}^L) - \beta \hat{y}_j^L \kappa^L$. Since the producer's participation constraint holds at $-\hat{q}^P + \beta \hat{y}_j^P \kappa^L = 0$, the producer realizes a positive surplus when trading with a high-type consumer, since $-\hat{q}^P + \beta \hat{y}_j^P \kappa^H > 0$. A separating offer, on the other hand, guarantees zero surplus for the producer, as established in Proposition A. Hence, for $\pi \rightarrow 0$ it follows that $\hat{S}_c^{sep}(a; \kappa^H) > \hat{S}_c^{pool}(a; \kappa^H)$. Consequentially, there exists a $\hat{\pi} \in (0, 1)$, where for $\pi > \hat{\pi}$, the equilibrium is pooling, and for $\pi < \hat{\pi}$, the equilibrium is separating. ■

H. Proof of Proposition F

The proof proceeds in three steps starting with the solution to the DM bargaining game in *part (a)*. In *part (b)*, we determine the threshold asset supply, $\check{a} \in (0, a^*)$, followed by the characterization of the undefeated equilibrium in *part (c)*.

Part (a): Let us start with the separating equilibrium. A χ -type consumer entering the DM with a units of the information-sensitive asset and m units of fiat money solves:

$$(q^\chi, y_S^\chi, y_C^\chi, l^\chi, d^\chi) \in \arg \max_{q, y_S, y_C, l, d} u(q) - \beta y_S \kappa^\chi - \beta l - \beta \phi d, \quad (\text{H.1})$$

$$\text{s.t. } -q + \beta y_S \kappa^\chi + \beta l + \beta \phi d \geq 0, \quad (\text{H.2})$$

$$\text{s.t. } -l + y_C \kappa^\chi \geq 0, \quad (\text{H.3})$$

$$\text{s.t. } -y_S - y_C + a \geq 0, \quad (\text{H.4})$$

$$\text{s.t. } -d + m \geq 0, \quad (\text{H.5})$$

$$\text{s.t. } u(q^L) - \beta[y_S^L \kappa^L + y_C^L \kappa^L + \phi d^L] \geq u(q^H) - \beta[y_S^H \kappa^L + y_C^H \kappa^L + \phi d^H], \quad (\text{H.6})$$

$$\text{s.t. } u(q^H) - \beta[y_S^H \kappa^H + y_C^H \kappa^H + \phi d^H] \geq u(q^L) - \beta[y_S^L \kappa^H + y_C^L \kappa^H + \phi d^L], \quad (\text{H.7})$$

where (H.2) represents the producer's participation constraint, (H.3) the collateral constraint, (H.4)-(H.5) the feasibility constraints, and (H.6)-(H.7) the incentive-compatibility constraints. The

solution to the κ^L -consumer's problem is:

$$q^L = \min\{q^*, \beta(a\kappa^L + \phi m)\}, \quad (\text{H.8})$$

$$y_j^L = \min\{[q^* - \beta\phi d^L]/\beta\kappa^L, a\}, \quad (\text{H.9})$$

$$d^L = \min\{[q^* - \beta y_j^L \kappa^L]/\beta\phi, m\}, \quad (\text{H.10})$$

$$l^L = y_c^L \kappa^L. \quad (\text{H.11})$$

The solution to the κ^H -consumer's problem, on the other hand, is:

$$q^H = q^* = \beta\phi d^H, \quad (\text{H.12})$$

$$y_j^H = 0, \quad (\text{H.13})$$

if $\beta\phi m \geq q^*$, and:

$$y_j^H = [q^H - \beta\phi m]/\beta\kappa^H, \quad (\text{H.14})$$

$$u(q^L) - q^L = u(q^H) - q^H + \left[1 - \frac{\kappa^L}{\kappa^H}\right][q^H - \beta\phi m], \quad (\text{H.15})$$

if $\beta\phi m < q^*$, where $q^L = \min\{q^*, \beta(a\kappa^L + \phi m)\}$, $q^H \leq q^L$ (holding with equality iff $\beta\phi m \geq q^*$), and $y_j^H \in [0, a]$. In words, there exists a pecking-order theory of payment choices in the separating equilibrium according to which κ^H -consumers use fiat money first and only sell/pledge assets if first-best consumption cannot be obtained with fiat money alone, i.e., if $\beta\phi m < q^*$.

Let us now consider the pooling equilibrium. A κ^H -consumer solves:

$$(\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l}, \bar{d}) \in \arg \max_{q, y_S, y_C, l, d} u(q) - \beta y_S \kappa^H - \beta y_C f - \beta\phi d, \quad (\text{H.16})$$

$$\text{s.t. } -q + \beta y_S R + \beta l + \beta\phi d \geq 0, \quad (\text{H.17})$$

$$\text{s.t. } -l + y_C[\pi f + (1 - \pi)\kappa^L] \geq 0, \quad (\text{H.18})$$

$$\text{s.t. } u(q) - \beta y_S \kappa^L - \beta y_C \kappa^L - \beta\phi d \geq u(q^L) - q^L, \quad (\text{H.19})$$

$$\text{s.t. } -y_S - y_C + a \geq 0, \quad (\text{H.20})$$

$$\text{s.t. } -d + m \geq 0. \quad (\text{H.21})$$

The solution to (H.16) is such that:

- (i) If $\beta(\phi\bar{d} + \bar{y}_C\kappa^L) \geq q^*$, then $\bar{q} = q^*$, $\bar{y}_S = 0$, $\bar{y}_C = [q^* - \beta\phi\bar{d}]/\beta\kappa^L$, $\bar{l} = \bar{y}_C\kappa^L$, and $\bar{d} = [q^* - \beta\bar{y}_C\kappa^L]/\beta\phi$;
- (ii) If $\beta(\phi\bar{d} + \bar{y}_C[\pi f + (1 - \pi)\kappa^L]) \in (\tilde{q}, q^*)$, then $\bar{q} \in (\tilde{q}, q^*)$, $\bar{y}_S = 0$, $\bar{y}_C = a$, $\bar{l} = \bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ with $f \in (\kappa^L, \kappa^H)$, and $\bar{d} = m$; and

(iii) If $\beta(\phi\bar{d} + \bar{y}_c[\pi\kappa^H + (1 - \pi)\kappa^L]) \in (0, \tilde{q})$, then $\bar{q} \in (0, \tilde{q})$, $\bar{y}_j = a - \bar{y}_{-j}$, $\bar{l} = \bar{y}_c R$, and $\bar{d} = m$;

where $\tilde{q}$ solves $u'(\tilde{q}) = \kappa^H/R$.

Part (b) Taking into account the solutions in *part (a)*, let us now determine the κ^H -consumer's optimal payment strategy as a function of her portfolio, (m, a) , and thus the threshold asset supply $\tilde{a} \in (0, a^*)$. Relying on the insights of *part (a)* in Appendix D identifying the single-crossing property of $S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a; \kappa^H)$ and $S_{c,f = \kappa^H}^{pool}(a; \kappa^H)$, the threshold asset supply, $\tilde{a}$, solves:

$$\begin{aligned} S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a, m; \kappa^H) &\equiv u[\beta(\phi m + \tilde{a}[\pi f + (1 - \pi)\kappa^L])] - \beta(\phi m + \tilde{a}f) \\ &= u(\tilde{q}) - \tilde{q}\kappa^H/R \equiv S_{c,f = \kappa^H}^{pool}(a, m; \kappa^H), \end{aligned} \quad (\text{H.22})$$

suggesting:

$$S_{c,f \in [\kappa^L, \kappa^H]}^{pool}(a, m; \kappa^H) - S_{c,f = \kappa^H}^{pool}(a, m; \kappa^H) \begin{cases} > 0 & \text{if } a > \tilde{a} \\ = 0 & \text{if } a = \tilde{a} \\ < 0 & \text{if } a < \tilde{a}. \end{cases} \quad (\text{H.23})$$

Using $\tilde{q} = \beta(aR + \phi m)$ and totally differentiating (H.22) with respect to m yields:

$$\frac{\partial \tilde{a}}{\partial m} = \frac{\phi[1 - u'(\tilde{q})]}{[\pi f + (1 - \pi)\kappa^L][u'(\tilde{q}) - f/[\pi f + (1 - \pi)\kappa^L]]} < 0, \quad (\text{H.24})$$

given $u'(\tilde{q}) - f/[\pi f + (1 - \pi)\kappa^L] > 0$ for $f < \kappa^H$ at $a = \tilde{a}$.

Part (c): Combining the insights in *part (a)* and *part (b)* yields the equilibrium allocations in Proposition F, and thus the undefeated equilibrium. ■

I. Proof of Lemma A

The proof follows directly from Proposition F. Plugging the equilibrium surpluses $S_c(a, m; \kappa^H) = u(\bar{q}) - \beta(\phi m + a f)$ and $S_c(a, m; \kappa^L) = u(\bar{q}) - \beta(\phi m + a \kappa^L)$ with $\bar{q} = \beta(\phi\bar{d} + \bar{y}_c[\pi f + (1 - \pi)\kappa^L])$ into the objective function (32) and taking the derivative with respect to a' and m' yields the equilibrium prices (33) and (34). ■

J. Proof of Proposition G

Plugging $i = \gamma/\beta - 1$ into (33) in Lemma A and incorporating the solutions in Proposition F yields the equilibrium allocations in Proposition G. ■

Appendix: Supplementary Material

K. Screening Game

The bargaining game has the structure of a screening game, i.e., the producer, uninformed about the consumer's type, $\chi = \{L, H\}$, moves first and makes an offer.²⁰ Precisely, the producer offers a menu of contracts $\{(q^L, y_S^L, y_C^L, l^L), (q^H, y_S^H, y_C^H, l^H)\}$ to maximize her expected match surplus according to:

$$(q^\chi, y_S^\chi, y_C^\chi, l^\chi) \in \arg \max_{q^\chi, y_S^\chi, y_C^\chi, l^\chi} \pi[-q^H + \beta y_S^H \kappa^H + \beta l^H] + (1 - \pi)[-q^L + \beta y_S^L \kappa^L + \beta l^L], \quad (\text{K.1})$$

subject to:

$$\text{s.t. } u(q^\chi) - \beta y_S^\chi \kappa^\chi - \beta y_C^\chi f^\chi \geq 0, \quad (\text{K.2})$$

$$\text{s.t. } y_C^\chi \mathbb{E}_\lambda[\kappa^\chi] - l^\chi \geq 0, \quad (\text{K.3})$$

$$\text{s.t. } a - y_S^\chi - y_C^\chi \geq 0, \quad (\text{K.4})$$

$$\text{s.t. } q^L, q^H \geq 0, \quad (\text{K.5})$$

$$\text{s.t. } u(q^H) - \beta y_S^H \kappa^H - \beta y_C^H \kappa^H \geq u(q^L) - \beta y_S^L \kappa^H - \beta y_C^L \kappa^H, \quad (\text{K.6})$$

$$\text{s.t. } u(q^L) - \beta y_S^L \kappa^L - \beta y_C^L \kappa^L \geq u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L. \quad (\text{K.7})$$

The inequality (K.2) represents the consumers' participation constraints, (K.3) the collateral constraint, and (K.4)-(K.5) the feasibility constraints. Lastly, the type-specific incentive-compatibility constraints are captured in (K.6) and (K.7). Let us first determine the separating equilibrium, i.e., the equilibrium with $(q^L, y_S^L, y_C^L, l^L) \neq (q^H, y_S^H, y_C^H, l^H)$.

Lemma K.1. The unique separating equilibrium solving (K.1) satisfies:

$$(i) \ u(q^H) - \beta y_S^H \kappa^H - \beta l^H = 0;$$

$$(ii) \ u(q^L) - \beta y_S^L \kappa^L - \beta l^L = u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L; \text{ and}$$

$$(iii) \ y_j^H \leq y_j^L \text{ and } q^H \leq q^L,$$

with $y_C^L \kappa^L = l^L$ and $y_C^H \kappa^H = l^H$.

Proof of Lemma K.1. *The proof proceeds step by step. (ii) Proof by contradiction. Suppose $u(q^L) - \beta y_S^L \kappa^L - \beta l^L > u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L$. Given $\kappa^H - \kappa^L > 0$, this would imply $u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L > u(q^H) - \beta y_S^H \kappa^H - \beta l^H$. Since the participation constraint states $u(q^H) - \beta y_S^H \kappa^H - \beta l^H = 0$, this would imply $u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L > 0$, which contradicts (K.2).*

²⁰Note that in the absence of information frictions, a consumer's incentive to carry assets into the DM is eliminated if producers set the terms of trade unilaterally. Nonexistence of a monetary equilibrium would be the consequence.

$\beta l^H \geq 0$, this would imply $u(q^L) - \beta y_S^L \kappa^L - \beta l^L > 0$, and thus $q^L > 0$. By choosing a smaller q^L , the producer could increase her surplus satisfying all the constraints. In fact, a decrease in q^L would satisfy (K.7) and relax (K.6). Thus, a lower q^L is feasible and advantageous for the producer, suggesting that (K.7) must be binding in equilibrium. (iii) follows from (ii) by plugging (ii) into (K.6) yielding $(y_S^L - y_S^H) \kappa^H \geq (y_S^L - y_S^H) \kappa^L$, suggesting $y_j^H \leq y_j^L$. Plugging $y_j^H \leq y_j^L$ into (ii) suggests $q^H \leq q^L$. Lastly, (i) is proven by contradiction. Suppose $u(q^H) - \beta y_S^H \kappa^H - \beta l^H > 0$, implying $q^H > 0$. Since $q^H \leq q^L$, this would imply $u(q^L) - \beta y_S^L \kappa^L - \beta l^L \geq u(q^H) - \beta y_S^H \kappa^L - \beta y_C^H \kappa^L \geq u(q^H) - \beta y_S^H \kappa^H - \beta l^H$, suggesting $u(q^L) - \beta y_S^L \kappa^L - \beta l^L > 0$. In said scenario, the producer could increase her surplus by offering an alternative menu of contracts, q'^L and q'^H , with $q'^L < q^L$ and $q'^H < q^H$ satisfying all the constraints. Thus, q^H and q^L can not be the solution to the problem, and thus $u(q^H) - \beta y_S^H \kappa^H - \beta l^H = 0$ needs to hold in equilibrium. ■

First and foremost, the results in Lemma K.1 show that in the separating equilibrium allocations are equivalent for collateralized credit and asset sales, analog to the previously studied signaling game. However, in the screening game, given (i), producers leave no surplus to the κ^H -consumers. κ^L -consumers, on the other hand, extract a surplus proportional to the amount of assets sold/ pledged by the κ^H -consumer, i.e., $u(q^L) - \beta y_S^L \kappa^L - \beta l^L = \beta y_j^H (\kappa^H - \kappa^L)$ in (ii). Lastly, according to (iii), κ^H -consumers transfer fewer assets and consume a lower quantity of DM goods than κ^L -consumer in equilibrium. Plugging (i) and (ii) in Lemma K.1 into (K.1) yields the updated producer's problem:

$$(q^L, q^H) \in \arg \max \pi [-q^H + u(q^H)] + (1 - \pi) \left[-q^L + u(q^L) - u(q^H) \frac{\kappa^H - \kappa^L}{\kappa^H} \right], \quad (\text{K.8})$$

subject to $a - y_j^x \geq 0$ and $q^L - q^H \geq 0$. Proposition K.2 summarizes the equilibrium allocations.

Proposition K.2. (Separating equilibrium - screening) The unique separating equilibrium solving (K.8) satisfies:

$$q^L = \min\{q^*, \beta a \kappa^L\}, \quad (\text{K.9})$$

and:

$$q^H = \begin{cases} = 0 & \text{if } \pi < (1 - \pi) \frac{\kappa^H - \kappa^L}{\kappa^H} \\ > 0 & \text{if } \pi > (1 - \pi) \frac{\kappa^H - \kappa^L}{\kappa^H}. \end{cases} \quad (\text{K.10})$$

Proof of Proposition K.2. Taking the derivative of (K.8) with respect to q^L yields $u'(q^L) - 1 \geq 0$, holding with equality if $a > a^*$, where a^* solves $q^* = \beta a^* \kappa^L$. Taking the derivative of (K.8) with respect to q^H yields $u'(q^H) [\pi - (1 - \pi) \frac{\kappa^H - \kappa^L}{\kappa^H}] = \pi$, and thus (K.10). ■

The results in Proposition K.2 show that the consumption of the κ^L -consumer is first-best if the feasibility constraint, (K.4), does not bind. Consumption of the κ^H -consumer, on the other

hand, depends on the distance between the returns κ^H and κ^L , as well as the distribution of types, π .

Having determined the allocations in the separating equilibrium, let us now determine the pooling equilibrium, i.e., the equilibrium with $(q^L, y_S^L, y_C^L, l^L) = (q^H, y_S^H, y_C^H, l^H) = (\bar{q}, \bar{y}_S, \bar{y}_C, \bar{l})$. Proposition K.3 summarizes the equilibrium allocations.

Proposition K.3. (Pooling equilibrium - screening) The unique pooling equilibrium solving (K.1) satisfies:

$$\bar{q} = u'^{-1}(f/[\pi f + (1 - \pi)\kappa^L]) \in (0, q^*], \quad (\text{K.11})$$

with $\bar{l} = \bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ and $f \in [\kappa^L, \kappa^H]$.

Proof of Proposition K.3. *The proof proceeds in two steps distinguishing between an asset sale and a collateralized credit. In the event of an asset sale, a producer maximizes $-\bar{q} + \beta\bar{y}_S R$ subject to $u(\bar{q}) - \beta\bar{y}_S \kappa^H \geq 0$, $a - \bar{y}_S \geq 0$, and $\bar{q} \geq 0$, where $u(\bar{q}) - \beta\bar{y}_S \kappa^H = 0$ in equilibrium. Plugging into the objective function and taking the derivative with respect to $\bar{q}$ yields $u'(\bar{q}) = \kappa^H/R > 1$, and thus $\bar{q} = u'^{-1}(\kappa^H/R) \in (0, q^*)$. In the event of a collateralized credit, the producer maximizes $-\bar{q} + \beta\bar{y}_C[\pi f + (1 - \pi)\kappa^L]$ subject to $u(\bar{q}) - \beta\bar{y}_C f = 0$ with $f \in [\kappa^L, \kappa^H]$, $a - \bar{y}_C \geq 0$, and $\bar{q} \geq 0$. Plugging into the objective function and taking the derivative with respect to $\bar{q}$ yields $u'(\bar{q}) - f/[\pi f + (1 - \pi)\kappa^L] \geq 0$, and thus $\bar{q} = u'^{-1}(f/[\pi f + (1 - \pi)\kappa^L]) \in (0, q^*]$ with $\bar{q} = q^*$ if $a \geq a^*$ where a^* solves $q^* = \beta a^* \kappa^L$. ■*

The equilibrium allocations in Proposition K.3 are analog to the pooling equilibrium allocations in the corresponding signaling game discussed in Proposition B showing non-equivalence of collateralized credit and asset sales.